From Discipline to Human Capital Formation: A Hybrid Multi-Stage Econometric Machine Learning Study of the Association between Disciplinary Actions and Learning Outcomes in Children Under 5

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Abstract

The early childhood period is a crucial stage that shapes foundational learning, as it is during this time that the home environment and caregiving practices have a significant influence on children's cognitive development. Despite evidence on the harmful effects of violent and harsh disciplinary actions, such practices still remain widespread in Pakistan. Thus, this study examines the relationship between household disciplinary practices and learning outcomes among children under the age of 5 in Punjab, Pakistan using an innovative hybrid multi-stage model combining econometrics with machine learning . Using data from MICS 2017-18, we construct an ordered logit model with a generalized variant for robustness. The results show that non-violent disciplinary practices, such as explaining behavior to a child and redirecting a child's attention, are positively associated with learning, while psychological aggression and physical punishment are linked to significantly poorer learning outcomes. In addition, we created a Random Forest model that indicates that the caregiver's belief about the necessity of physical punishment is the strongest predictor of disciplinary practices using interpretative machine learning techniques like permutation importance. To ensure robustness against correlated features, we apply a Leave-One-Covariate-Out (LOCO) validation to the top two predictors, confirming their unique explanatory power through targeted ablation. These findings then underscore the importance of shifting parental norms toward non-violent discipline to strengthen early learning and human capital formation.

Introduction

Developing an individual's potential and a country's human capital depends on giving children the best possible start in life (World Bank, 2024). The first five years of life are the fastest period of human growth and development as 90 percent of a person's brain development occurs during this period (World Bank, 2024; Saracoğlu & Karaoğlan, 2016). This neurobiological base is highly influenced by the early environment, so the home becomes one of the main places of forming or degrading human capital. During these early years, the body passes through a maturation process in which there is a sequential development of perceptual, sensory, motor, cognitive and socio-emotional skills (UNICEF, 2022). If this sensitive stage is

characterized by any negative experience like violent disciplinary methods, it poses a direct threat to the developmental pathways, this undermines the competence of caregivers and the cognitive development of a child (UNICEF, 2019). Thus this paper sheds light on an urgent matter that Pakistan needs to prioritize, early childhood development (ECD). Since this is the cornerstone of the Human Capital Index; it is impossible to repair the damage to cognitive development caused by the lack of protection and stimulation of a child under the age of five, and the effects on the national productivity of the nation are permanent (World Bank, 2024). However, understanding disciplinary practices is inherently complex due to their multifaceted nature-practices often co-occur, are culturally embedded, and interact with socioeconomic and child-specific factors in ways that resist simple causal interpretation. Such behavioral patterns cannot be adequately captured through conventional analytical models alone.This paper employs a hybrid econometric-machine learning approach to quantify and map these complex disciplinary dynamics. It seeks to investigate the relationship between household disciplinary practices and their effect on learning outcomes and mental health of children under the age of five, including the predictors of such practices. It claims that disciplinary strategies that may be embedded in cultural norms, and that parental belief systems have a strong relation with the underlying capabilities that influence the ultimate academic achievement, the mental health and, eventually, the human capital of a nation.

Literature Review

Disciplinary Actions and Prevalence

Disciplinary measures are a fundamental element of bringing up children and it involves a set of strategies that caregivers use to redirect, correct, or control the behavior of children. These measures are formulated in general terms as the reaction to misbehavior of children to ensure socialization and compliance with societal norms (Valencia et al., 2025). Non-violent or positive discipline is intended to educate and correct children, it does not cause damage but focuses on emotion control and behavior management and is associated with a healthier socioemotional growth (Cuartas et al., 2019). Violent discipline, on the other hand, is classified into two major forms. Physical punishment (corporal punishment) refers to any form of punishment where physical force is applied and is intended to produce some amount of pain or

discomfort (WHO, 2021). Psychological aggression is an infliction of psychological pain, which can be brought about by embarrassing, mocking, humiliating, or making someone feel worthless (Cuartas et al., 2019).Importantly, non-violent techniques are normally used in conjunction with violent techniques and children who received non-violent discipline continue to be a minority in the world (UNICEF, 2010).

Global Context

Across the world, disciplinary practices differ remarkably in terms of their prevalence, their way of manifestation and this is influenced by cultural, socioeconomic and legal factors. Worldwide, nearly 1.6 billion children between ages one and fourteen are subjected to violent discipline by their caregivers with almost 400 million children being below the age of five (UNICEF, 2017). In most of the low- and middle-income countries (LMICs), corporal punishment is still widely practiced, and an estimated 1.2 billion children experience physical punishment each year (WHO, 2021). Even more prominent is psychological aggression, which affects more than 75 percent of children between 2 and 4 years of age all over the world (UNICEF, 2010). Non-violent discipline is rather less common as a sole method, and only a minor part of households use it without violence (UNICEF, 2010). These contextual realities of the world highlight normative legitimacy of severe punishment in the light of increasing evidence of its developmental harm.

Disciplinary Action in the Context of Pakistan

In Pakistan, this global trend is also expressed in disciplinary practices but the use of practice is conditioned by specific cultural and legal peculiarities. Corporal punishment is common in the country, with about 8.9 million children between the ages of 2 and 4 being its target. Pakistan also ranks the fourth most exposed to physical aggression among the 49 low and middle-income countries (Ashraf and Holden, 2022). There are significant regional differences in physical punishment with a prevalence rate of 42 percent in Gilgit Baltistan, more than 60 percent in Punjab and Sindh (Ashraf and Holden, 2022). Even more widespread is psychological aggression which affects over 70 percent of children in Gilgit-Baltistan, Punjab, and Sindh (Corporal Punishment of Children in Pakistan, 2024). In the same way, non-violent discipline is still not widely used in Pakistan. According to national survey data, only a minority of children

(less than 10 percent) were brought up by non-violent means in all three provinces. Violent discipline becomes typically sustained by cultural norms, beliefs of parents, and religious factors which frequently justify or reinforce physical and psychological aggression (Ullah et al., 2023; Ashraf and Holden, 2022). Legal changes, like the 2021 Islamabad Capital Territory bill against corporal punishment in schools and care facilities, are a step forward, but they are not enough to stop physical punishment in the household and this weakens their effect (Bess, 2022). This creates a paradox in which a practice that has proven to have developmental risks being legally acceptable and even within the social fabric in the inner world where young children spend a huge amount of their time.

The Disciplinary Action Paradox

The disciplinary action paradox is the normative acceptability of this action in contrast to the existing harms of disciplinary action that are well documented. Although corporal punishment and psychological aggression are often viewed as effective methods for securing immediate compliance, they should be distinguished from their long-term effects, as caregivers may focus solely on short-term obedience (Gershoff, 2002; Cuartas et al., 2019). This paradox is also particularly visible in Pakistan, where severe disciplinary practices are culturally embedded and contribute to the challenges in education, such as school dropouts and psychological distress among children (Bess, 2022; Mukhi, 2025). Although the Pakistan Penal Code Section 89 established a historical precedent for guardians to give physical punishment under the banner of good faith, legal changes, such as those through the ICT Child Protection Act 2018 and provincial legislation in Sindh have indicated the gradual creation of a coherent legal structure against the use of physical punishment. However, the paradox of disciplinary action still exists, and the normative acceptance of these forms of discipline by the majority of Pakistani parents remains strong despite the existence of evidence demonstrating the negative impact of physical discipline on development (Aslam et al., 2021).

Most Affected Age Group

Age-specific vulnerability is critical in understanding the effects of disciplinary practices. The intersection of developmental, neurological, and psychological pathways in children under five exposes them to the adverse effects of harsh discipline at a comparatively high rate. Their

survival and well-being rely solely on their caregivers. A punitive caregiver is not just someone who corrects the child when they are wrong, but a threat to the very essence of safety and security of the child.

As these children are undergoing rapid neurological growth, any harmful disciplinary practice imposed on them can lead to structural brain transformations, which damage moral judgment and empathy (Whittle et al., 2022). Violence exposes the brain to toxic stress at an early age and disrupts the neural pathways, in particular, the prefrontal cortex and amygdala, which are crucial in language, executive functioning, and emotional regulation, which form the basis of learning and social competence (Illachura et al., 2024; Pace et al., 2018; Gershoff, 2016). For children under five, these disruptions occur just as foundational literacy and numeracy development are beginning to emerge, including letter recognition and number identification. For 3-5 year aged children, a study analyzed data from 35 MICS countries and found that children from low human development index (HDI) countries have a higher impact of physical punishment as reflected by their low child development scores as compared to those living in high and medium HDI countries (Pace et al., 2018).

Predictors

Research from high and middle-income countries consistently identifies a core set of predictors for harsh discipline, centering on parental background, cognitive-affective processes, and contextual stressors.

Parental Factors

Beyond personal history, background, parental cognitive and affective states are powerful proximal determinants. The Social Information Processing (SIP) model provides a useful framework where maternal cognitive risk factors, such as low frustration tolerance and an external locus of control, significantly predict abuse potential and maladaptive discipline (McElroy & Rodriguez 2008). Similarly, Ateah and Durrant (2005) identify that a mother's anger in response to misbehavior and her approval of physical punishment were key predictors of its use. This is complemented by Rodriguez (2010), who highlights the role of contextual stress and avoidant coping strategies in increasing abuse risk. Further supporting the role of parental attitudes and affect, Jackson et al. (1999), in a nationally representative U.S. sample, found that

anger mismanagement and attitudes that devalue children predicted more favorable attitudes toward physical discipline, which in turn influenced harsher practices.

Socio-economic Factors

The influence of socioeconomic context is also evident in high-income countries. Dietz (2000), analyzing a U.S. national sample, found that parents with fewer resources (lower income, education) and those socialized into violent subcultures were more likely to use severe corporal punishment. Evans (2015) confirmed that low-income U.S. families with more demographic risks (e.g., single parenthood, receipt of public assistance, unemployment, teenage parenthood, and lack of a high school diploma or GED) were more prone to choosing corporal punishment over non-violent strategies.

Child Characteristics as a Predictor

Factors on child level are also significant predictors of disciplinary action. A German study by Engfer and Schneewind (1982) identified a child perceived as "difficult" as a primary predictor of harsh maternal punishment. Parental responses also depend on the temperament or behavioral phenotype of the child. Some children exhibit “punishments insensitivity construct” which can be identified in preschoolers, as the child shows reduced responsiveness to common punitive corrections (Nichols et al., 2014). Parents of children with such atypical, recalcitrant behavior can progress to more extreme forms of punishment, and a negative cycle of interaction is enforced.

Normalized Intra-Familial Violence

Other major themes are the normalization of family aggression and the intergenerational transmission of violence. Rodriguez and Sutherland (1999) found that a parent’s own history of experiencing a discipline technique was the strongest predictor of its use, especially when the child was perceived as culpable. This link between childhood experiences and future parenting is further discussed by Rodriguez and Price (2004), who found that young adults who experienced harsh discipline and believed they deserved it, showed higher abuse potential and intended to use harsher discipline themselves. Ahmed et al. (2025) demonstrate that a mother's exposure to spousal violence directly predicted her use of aggressive disciplinary practices, providing a

striking example of the connection between violence against women and violence against children. All of this research shows that disciplinary measures are part of a larger framework of economic hardship and accepted intra-familial violence in many LMICs. In a national child welfare sample, mothers with a history of domestic violence (DV) victimization reported significantly higher use of aggressive and neglectful disciplinary practices toward their children. After controlling for covariates, recent DV was associated with increased physical aggression and neglect, while both recent and past DV were linked to greater psychological aggression (Kelleher et al., 2008).

Pakistan-Specific Investigations

The limited research from Pakistan is in accordance with findings in other countries around the world; however, it also provides critical information surrounding the cultural and social factors that perpetuate harsh discipline. In Lahore, physical and psychological disciplinary action among adolescents was significantly associated with having an authoritarian parenting style, a larger family size, and living in a joint family setup (Rizvi & Najam, 2019). Therefore, it appears that traditional hierarchical family structures may use severe discipline to enforce their strict authority. Parental rejection is the strongest predictor of child abuse, states Malik (2010), and other significant predictors include large family size and low maternal education. This study's noteworthy findings that emotional abuse and neglect were more prevalent than physical abuse defied preconceived notions and emphasized the necessity of a broader definition of maltreatment in Pakistan.

Impact of Disciplinary Action

The use of discipline practices has both long-term and short-term effects on the learning results and mental health of children. Corporal punishment is also associated with poorer cognitive ability, and less executive functioning and impaired self-regulation, which adversely affect school preparedness and academic performance (Cuartas, 2022; Gershoff and Grogan-Kaylor, 2016). Corporal punishment was also associated with a reduction in creative skill, reduced self-confidence, and an increase in anxiety in Pakistani children (Aslam et al., 2021). Psychological aggression also leads to internalizing of issues like anxiety, depression, and low self-esteem, which adversely impact the interaction with school and social adaptation (UNICEF,

2010; Muhammad and Zakar, 2016). These young children lack the cognitive maturity to contextualize punishment. They are more likely to internalize the action as a global statement of their worth rather than a correction of a specific behavior hence, this fosters shame and low self-esteem.

Long-term Consequences

Gershoff's (2002) meta-analysis of 88 studies provides one of the most comprehensive assessments of the developmental impacts of corporal punishment on children of different ages, finding that while corporal punishment does significantly increase immediate compliance, chances of delinquent behavior also increase, while the parent-child relationship worsens. The study also found that the effects of corporal punishment carry onto adulthood, with the person seeing greater aggression and antisocial behavior having an increased likelihood of abusing their own spouse or child. Aggressive parental strategies, along with low levels of parental warmth have also been linked to criminal behavior later in adulthood, with harsh parenting at age 10 being linked to externalizing behavior in adulthood (Gershoff, 2002; Taillieu & Brownridge, 2015).

Importance of Human Capital Investment in Early Stage

The connection of disciplinary actions to human capital is demonstrated by early intervention programs. One such program was the *Preparing for Life* trial, which offered intensive parenting support throughout pregnancy to five years old; this led to major improvements in children's cognitive skills and in school achievement tests in reading and math at the age of nine. The mediation analysis showed that 35-38% of this sustained effect was an outcome of child-related improvements in early parental beliefs and investments (Doyle, 2022). This highlights the idea that the change of parental disciplinary practices and beliefs is not only about reducing harm but also about constructing cognitive capital, which is the major driving force of productivity in the future. Conversely, when the investment in early childhood education is low, as in the case of Turkey, educational attainment and a less skilled workforce will be lower in the long-term (Saracoğlu & Karaoğlan, 2016). In Pakistan, the lack of these early learning basics, enhanced by violent discipline, forms a human capital crisis in which children achieve a

small fraction of the potential adult productivity, impairing its human capital and economic development over the long term (UNICEF, 2022; World Bank, 2024).

Problem Statement

Despite this extensive literature on disciplinary strategies and their impacts, substantial gaps exist in understanding this issue especially in the Pakistani context. First, there is a limited focus in literature on the impact of disciplinary action on children under five, as most studies focus on older children and adolescents. However, gaining this insight is crucial as early childhood represents a critical stage in the life cycle approach of human capital formation during which cognitive, emotional, and socio-behavioral foundations are formed (World Bank, 2024). There is also limited literature specifically focusing on the numeracy and literacy aspects of early childhood development despite it being an important factor as this can directly impact later education and overall well-being, therefore shaping the long-term human capital of Pakistan (World Bank, 2024). Moreover, even methodological gaps exist in studies focused on this topic because they look at disciplinary methods individually, instead of examining how parents use a combination of them in real life .There is also a limited use of predictive modeling in such research that can effectively address the complex interactions between child and household factors that shape disciplinary practices and their outcomes.

Therefore, to address these methodological and literature gaps our study uses the Multiple Indicator Cluster Survey (MICS) to examine the impact of disciplinary practices on children under five in Pakistan. In order to undertake this we adopt a hybrid multi-stage methodological approach, combining advanced econometric techniques, an ordered logit model with a generalized ordered logit variant for robustness checks. In addition, we employ a machine learning model, the random forest, for exploratory analysis to identify the strongest predictors of these disciplinary profiles from a comprehensive set of variables. To robustly validate the feature importance rankings derived from permutation importance, we employed a Leave-One-Covariate-Out (LOCO) approach on the top two predictors identified by the Random Forest model. This ensured that the importance of key predictors - such as caregivers' belief in the necessity of physical punishment - remained stable even in the presence of correlated features, reinforcing the reliability of our findings. This integrated approach then provides a holistic and

contextually nuanced understanding of how disciplinary practices prevail in Pakistani households. Thus, this study is guided by the following research questions:

Primary Research Question

How are different disciplinary practices, ranging from non-violent methods to severe corporal punishment, associated with foundational learning in early childhood?

Sub Question

What combination of child, caregiver, and household characteristics predicts the use of specific disciplinary actions by caregivers?

Sample and Descriptive Statistics

MICS Dataset

This paper uses the MICS 2017-18 dataset for Punjab, conducted by UNICEF in collaboration with the Punjab Bureau of Statistics. MICS is one of the largest sources of statistically sound and internationally comparable data on women and children worldwide. Most of the data was collected through face-to-face interviews and a majority of the data on child development was provided by mothers (Loizillon et al., 2017). The dataset includes information on a variety of areas, including health, early childhood development, caregiving practices and education. The survey uses a multi-stage, stratified cluster sampling approach to collect data, allowing for diverse representation from both rural and urban areas while also having a large sample size. Households were selected using a two-stage sampling design. In the first stage, census enumeration areas were selected within each strata, which was formed by dividing each district into rural and urban, with the probability being selected proportional to its size. In the second stage, household listings from the 2017 Census, provided by the Pakistan Bureau of Statistics, were used to draw a sample of 20 households from each selected enumeration area. The final sample included 53,840 households across 2,692 clusters (International Household Survey Network, 2021). The MICS dataset has standardised and detailed measures for measuring how children are disciplined across households, as well as measures for different kinds of early childhood development indicators, including questions on the child's literacy skills, both important features for this research. The MICS dataset has also been used in notable research

work on discipline and child development across the world, making it a credible and policy-relevant data source for this paper.

Sample

The sample for this study includes 15,590 observations of children between the ages of 36 and 60 months, with a mean age of 46.9 months, or around 3.9 years, and a relatively even distribution across the range. Observations for children older than 60 months were dropped so that results were consistent with the intended age range for the learning score indicator, as well as to avoid age-related measurement bias. The gender distribution is balanced, the sample being 49% female and 51% male, and 6% of the sample having a disability.

Looking at household characteristics, the sample is predominantly rural, with only 26.4% of children being from urban areas. Wealth distribution is also skewed (See Figure A1 in Appendix) and there are many more poorer households in the sample compared to richer households, which reflects the general socioeconomic composition of Punjab's population (Punjab Economic Report, 2015). Mother's education levels vary substantially as well but are predominantly low (See Figure A2 in Appendix), with around 46% of mothers reporting that they did not have a formal education or only had a preschool level education, while 26% of mothers reported a primary level education. Preschool attendance among children in the sample is also low, and only a third of the sample reported attending preschool, which further highlights the importance of the home environment in early childhood development. Table 1 shows a summary of the sample statistics.

Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Age (months)	15716	46.893	6.875	35	60
Female	15716	.486	.5	0	1
Disability	15716	.06	.237	0	1
Attended Preschool	15713	.334	.472	0	1
Mother's Education Level	15716	1.232	1.425	0	4
Wealth Index	15716	2.763	1.391	1	5
Urban	15716	.264	.441	0	1
Take Away Privileges	15701	.391	.488	0	1
Explain Behaviour	15699	.693	.461	0	1
Shook Child	15710	.649	.477	0	1
Shouted/Yelled at child	15710	.755	.43	0	1
Gave Something Else	15705	.436	.496	0	1
Spanked/Hit/Slapped	15707	.439	.496	0	1
Hit Bottom with Item	15701	.069	.253	0	1
Calling Dumb/Lazy/Stupid	15702	.318	.466	0	1
Hit Face/Head/Ears	15701	.507	.5	0	1
Hit Hand/Arm/Leg	15703	.381	.486	0	1
Beat as Hard as One Could	15702	.056	.23	0	1
Learning Score	15716	.814	1.135	0	3

Table 1: presents the descriptive statistics for the dataset, including the number of observations, mean, standard deviation and the minimum and maximum values for each variable

Disciplinary Actions

The spread of disciplinary action varies in the dataset, with non-physical disciplinary actions being the most common (See Figure 1). Explaining why the child's behaviour was wrong is one of the most commonly reported disciplinary actions, with 69.3% of caretakers saying that they used this method. Verbal disciplinary practices are also common as a majority of children were reported to have been shouted or yelled at. On the other hand, physical forms of discipline vary a lot, and practices like shaking a child are very common, showing that physical punishment remains common in early childhood. However, more severe forms of physical discipline, like hitting as hard as one could or hitting the child with an object on the bottom, are relatively rarer with only 5.6% and 6.9% of respondents using these methods.

Psychological aggression is also common, with around 75.5% of respondents stating that they had yelled or shouted at their child, while 31.8% of caretakers reported calling the child lazy, dumb or stupid, which also highlights how non-physical but still potentially harmful forms of

discipline are also prevalent. Overall, the figure shows how most children are exposed to multiple forms of discipline, including non-violent, psychological, and physical practices.

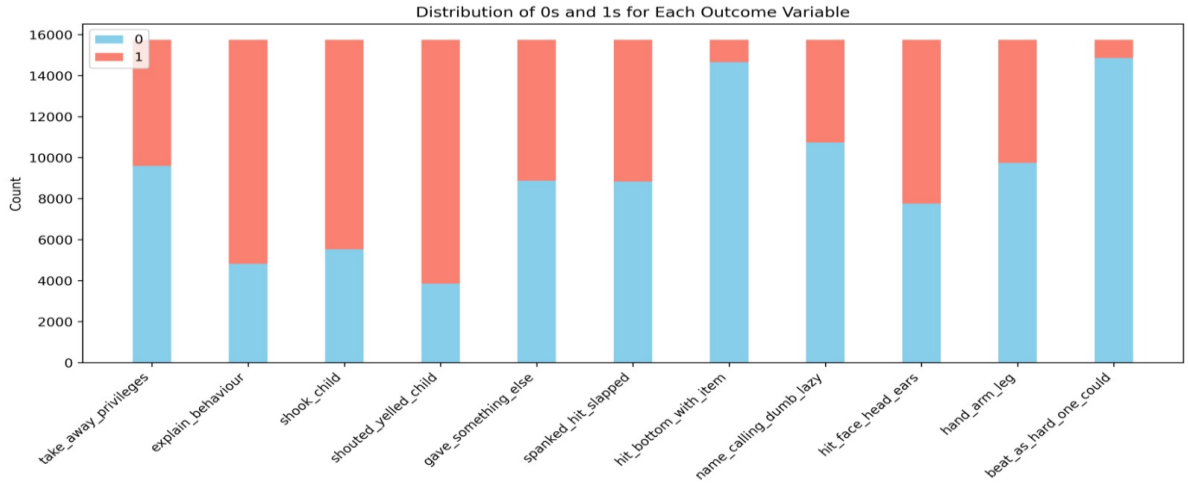


Figure 1. Overlaid bar chart depicting the proportion of households using each type of disciplinary practice.

Methodology

Dependant Variable (Learning Index)

Construction of validated composite indices is a common practice in educational assessment, as demonstrated in a recent work on literacy and numeracy tools by Fatimah et al. (2024). The direct measure of the core competencies, which is fundamental literacy and numeracy skills, as outlined in the national policy framework of Pakistan, is the learning index that is used in the present study. Based on the constructivist view of learning as hierarchical, the study's approach aligns with the principle that resource-constrained systems, much like Pakistan, must prioritize core 'gateway skills' (Salim, 2024). Summing up three important indicators on the UNICEF Early Childhood Development Index (ECDI) Literacy-Numeracy domain, EC6 (identify/ name at least ten letters of alphabet), EC7 (read at least four popular words), and EC8 (recognize/name all numbers 1- 10), this paper adheres to that principle. Also, an ordered scale (0-3) as a dependent variable would offer a more in-depth analysis of the relationships between factors and the levels of mastery because it is a more sensitive and informative scale.

According to the UNICEF ECDI robust framework, the EC6, EC7, and EC8 were not selected arbitrarily, a multi-year process involving literature synthesis, pilot testing in various countries, and psychometric analysis was conducted by experts. Although the index is based on the caregiver report, not direct child assessment, which is a typical characteristic of large-scale household surveys, after this rigorous testing, it was found to be internally consistent, culturally invariant and reliable proxy of foundational learning. The scoring system of ECDI stipulates that the scores of the literacy-numeracy domain will be calculated based on these three indicators which forms the precedent of integrating these indicators to form a valid composite index (Loizillon et al., 2017, p. 23). This is strongly evidenced in our sample by excellent internal reliability (Cronbach's $\alpha = 0.81$). Hence, the development of a composite index is a direct application and rational extension of the validated design of the instrument, moving from its binary outcome to a more granular measure of skill acquisition. Moreover, since this study involves analysis at the population level, application of such an index fits the purpose of the ECDI, as a concise, comparable tool for large-scale assessment.

Finally, the Learning Index was constructed by summing affirmative responses to the three ECDI items (EC6, EC7, and EC8), resulting in an ordinal variable with a range of 0-3, where 0 means no mastery of either one of the three foundational literacy and numeracy skills, 1 means mastery of any one of the three foundational literacy and numeracy skills, 2 means mastery of any two of the three foundational literacy and numeracy skills, 3 means mastery of all three foundational literacy and numeracy skills.

Independent Variables

The independent variables used for the econometric models are based on extensive literature review demonstrating their significant influence on child learning outcomes. The variables included wealth index, mother's education, area type, child gender, child age, attendance of school, child disability status. The justification derived by strong literature reference for each independent variable is attached in Appendix B.

Disciplinary actions variable

The independent variables for disciplinary practices were drawn from the Child Discipline module of MICS (UCD2A-UCD2K). In line with the survey's established conceptual framework for monitoring Sustainable Development Goal Indicator 16.2.1 (UNICEF, 2023), these items are often categorized into three domains: physical discipline (shook child, spanked/hit/slapped, hit face/head/ears, hit hand/arm/leg, hit bottom with item, beat as hard as one could), psychological aggression (shouted/yelled, called names), and non-violent discipline (took away privileges, explained behavior, gave something else to do).

While these categories provide a useful theoretical structure, we analyzed each of the 11 disciplinary actions individually in our econometric and machine learning model for several reasons. First, prior evidence suggests that specific behaviors may affect child outcomes through distinct pathways, and their associations can vary significantly even within a single domain (Gershoff et al., 2010). Second, caregivers often use multiple disciplinary actions simultaneously, supported by our data, hence analyzing them separately allows us to isolate the unique association of each action while controlling for the co-occurrence of others.

The psychometric analysis further supports the item level approach, whereby the pre-defined categories like physical, verbal, non-violent discipline shows moderate internal consistency (Cronbach's α: physical = 0.68, psychological = 0.42, non-violent = 0.58) which indicates that aggregation risks obscuring meaningful variation. There is a higher consistency of the 11 items when considered as a broader set ($\alpha = 0.77$). Aggregating the items into heterogeneous subscales would mask the unique relationships that specific behaviors have with learning outcomes.

Correlation and Multi-collinearity check

To assess potential multicollinearity among predictor variables, we computed a correlation matrix across all features used in the model. The results indicated low to moderate correlations, with the highest observed coefficient being 0.55 between the disciplinary actions "shouted/yelled at child" and "shook child." Given that this value falls within an acceptable range for moderate correlation and does not approach thresholds typically associated with harmful multicollinearity (e.g., >0.8), we conclude that multicollinearity is not a substantive concern in our model. This supports the reliability and stability of our coefficient estimates and reduces the risk of inflated

variance or biased interpretation in both econometric and machine learning results (See Figure B1 in Appendix B)

Models

For our analysis we used a hybrid multi-stage approach integrating advanced econometric and machine learning models. This methodological framework draws inspiration from Malhotra (2020), who advocates for the complementary use of conventional econometrics and machine learning to address complex, high-dimensional behavioral data where traditional methods may fall short-particularly in the context of relative importance analysis. In line with this approach, we first use econometric modeling to establish statistical significance, followed by machine learning techniques to uncover predictive patterns and rank variable importance in a robust, data-driven manner.A summary chart of the methodology is attached at the end of this section (See Figure 3)

Econometric Model

Ordered Logit

To address the primary research question of the study, we estimate an Ordered Logit regression model. The choice of this econometric model is based on the categorical nature of our dependent variable: the composite learning index, which takes on four discrete values (k = 0, 1, 2, 3), where higher scores represent greater mastery. This structure is fundamentally ordinal. An Ordinary Least Squares (OLS) regression would treat the distances between these categories as equal, which is not a valid assumption for an ordinal scale (Long, 1997; Wooldridge, 2010). Conversely, a Multinomial Logit would treat the categories as unordered, ignoring the inherent ranking and sacrificing statistical power (Cameron & Trivedi, 2005). The Ordered Logit model is therefore the appropriate specification for analyzing such graded outcome variables, and its use is well-established in educational and developmental research (Sesay et al., 2021)

The model is specified as follows.

Let y^*_i represent the unobserved, continuous learning propensity for child i:

$$\mathbf{y^*_i = \beta_0 + \beta' X_i + \varepsilon_i}$$

where X_i is a vector of covariates including wealth quintile (poor, second, middle, fourth, richest), area type (rural/urban), mother's education (none, primary, middle, higher secondary), child gender (male/female), child age, school program, child disability status, and eleven disciplinary practices: taking away privileges, explaining behavior, shaking the child, shouting or yelling at the child, giving the child something else to do, spanking/hitting/slapping the child, hitting the child's bottom with an item, calling the child dumb or lazy, hitting the child's face/head/ears, hitting the child's hand/arm/leg, and beating the child as hard as one could. β denotes the vector of coefficients, and ε_i, error term, follows a logistic distribution.

The observed categorical learning outcome y_i, taking values 0, 1, 2, or 3, is determined by the following threshold rules:

$$y_i = 0 \text{ if } y^*_i \leq \tau_0$$

$$y_i = 1 \text{ if } \tau_0 < y^*_i \leq \tau_1$$

$$y_i = 2 \text{ if } \tau_1 < y^*_i \leq \tau_2$$

$$y_i = 3 \text{ if } y^*_i > \tau_2$$

where τ_0, τ_1, and τ_2 are the estimated threshold (cut-point) parameters.

Estimation and Inference

Model parameters were estimated using maximum likelihood. To ensure robust inference, the analysis accounts for the complex survey design by applying individual sampling weights and clustering standard errors at the primary sampling unit level (HH1).

Interpretation

Because the raw ordered logit coefficients represent changes in log-odds, Average Marginal Effects (AMEs) are reported instead for simplicity. The AMEs provide an intuitive, probability-based interpretation of how each predictor affects the likelihood of a child being in each category of the Learning Index

Robust Extension of the Main Model

Generalized Ordered Logit

As a measure of robustness to complement the study's main model, additional statistical procedures are conducted following the primary Ordered Logit estimation. A major assumption of the Ordered logit is the proportional odds (parallel lines) assumption, which requires that each explanatory variable should be constant across all thresholds of the ordinal outcome. Given the skewed nature of the Learning Index distribution (See Figure 2) , the assumption is especially relevant in this current study.



Figure 3: The distribution of the Learning Index dependent variable

The Brant test is employed as a formal diagnostic to statistically assess whether proportional odds assumptions hold for each covariate (Brant, 1990). To address the potential violation of the assumption, a Generalized Ordered Logit model (Williams, 2006) is estimated as a robustness check. The model is implemented using the gologit2 command with the autofit option, this identifies the variables that violate the proportional odds assumption and then estimates a partial proportional odds model. This model allows only the identified coefficients to differ across thresholds, while the remaining variables retain proportional effects (Cyrenne & Chan, 2022).

The model is specified as follows:

$$\mathbf{P(Y_i > j) = \exp(\alpha_j + X_i\beta_j) / [1 + \exp(\alpha_j + X_i\beta_j)]}$$

for j = 0, 1, 2, 3

here α_j represents the threshold-specific intercepts and β_j represents a vector of coefficients that may vary across thresholds j. Estimation uses the same set of covariates and outcome variable (learning index) as the main model of Ordered Logit model and applies cluster-robust standard errors at the primary sampling unit level (HH1).

This procedural technique of first testing the proportional odds assumption using Brant test and then implementing a Generalized Ordered Logit model for robustness is an established best practice in the analysis or ordinal outcomes. For example, in recent research, Cyrenne and Chan (2022) adopt a similar sequential approach (Ordered Logit followed by Brant testing and Generalized Ordered Logit estimation) to examine determinants of university student success. This study provides a direct methodological precedent for the analytical framework employed in this study.

Random Forest

To answer the sub research question, as an exploratory analysis , we employ a random forest classifier for each disciplinary outcome to identify the most important predictors without imposing linear or parametric assumptions.

Studying disciplinary actions is methodologically challenging due to the complex, often non-linear, and high-dimensional nature of behavioral data (e.g., multiple co-occurring actions, interactions between household characteristics, child traits, and caregiver behaviors). While econometrics methods provide strong causal interpretations, they fail to fully capture intricate patterns of the real world. This study employs an innovative hybrid multi-stage methodological framework to answer the study's complex research questions by integrating machine learning for predictive and exploratory analysis with econometric modeling for statistical interpretation, thereby leveraging the complementary strengths of both analytical traditions (Varian, 2014).

Random Forest is selected as the primary machine learning technique due to several methodological advantages aligned with the characteristics of disciplinary data. The ensemble structure of the algorithm constructs multiple decision trees on bootstrapped samples and aggregates their predictions, providing robustness against overfitting and noise, something very

valuable when performing an analysis of self-reported behavioral data that are prone to measurement error. Unlike boosting methods that sequentially correct errors and can be sensitive to noisy labels, parallel tree construction in Random Forest adds to stability and reduces overfitting risk in imbalanced datasets (Breiman, 2001). Non-linear relationships and interaction effects are automatically handled through recursive partitioning without explicit specification, therefore making the method suitable for exploring complex behavioral phenomena. Furthermore because disciplinary practices frequently co-occur-we face a multi-label classification problem rather than a mutually exclusive one. Following established practices in machine learning literature (e.g., Ashwinkumar et al., 2024), we employ the standard One-vs-Rest approach, training 11 separate binary classifiers rather than forcing artificial mutual exclusivity. This method is specifically recommended for multi-label problems where outcomes naturally co-occur, allowing us to model each disciplinary practice independently while acknowledging their real-world simultaneity.

The most important results from the random forest are a computed list of feature importance, which will answer the study research question directly. To understand the importance of all the features and identify the strongest predictor in the model, permutation technique was used- a well-established method in interpretable machine learning that helps mitigate the "black-box" criticism often leveled at complex ML models (Molnar, 2020). Although random forest has an intrinsic feature importance measure through mean decrease in Gini impurity and entropy. These measures are typically biased towards variables with larger cardinality (Nembrini et al., 2018). Hence, a more reliable technique is permutation which assesses the contribution of each feature by measuring the decrease in model performance after randomly permuting its values (Altman et al., 2010).

Formally, permutation importance for feature *j* is defined as the difference between the model's baseline predictive performance and the performance obtained after randomly permuting the values of predictor X_j, while keeping all other predictors unchanged. Let *L(Y, f(X))* denote the model loss. The permutation importance is given by:

$$\mathbf{PI_j = L(Y, f(X^p_j)) - L(Y, f(X))}$$

where X^{p}_{j} represents the feature matrix in which predictor X_j has been randomly permuted. A larger increase in loss indicates greater importance of the feature.

The complete Implementation Pipeline

The methodological implementation follows a structured pipeline:

- Selection of Input Variables: Predictor variables were selected on the basis of a thorough literature review. Each variable and its rationale of inclusion is presented in a detailed table in Appendix C (See Table C) . Although Random Forest handles multi-collenarity well, a correlation matrix of all the input features is attached to Appendix C (Figure C1 and C2). Pairwise correlation analysis revealed generally low to moderate associations among input features, with the strongest correlation observed between weight and height variables (r>0.9), an expected relationship given their physiological interdependence. While this correlation is high, it does not pose a concern for multicollinearity within tree-based models such as Random Forest, which are inherently robust to correlated predictors. This supports the stability and interpretability of our feature importance rankings.
- Data Preprocessing: Missing values were addressed through median imputation so that the models become robust with the outliers.
- Class Balancing: For disciplinary outcomes exhibiting significant imbalance (like beating as hard as one could and hitting bottom with item), the Synthetic Minority Oversampling Technique (SMOTE) was applied.
- Model Training: Every single disciplinary action was trained with a one-vs-rest Random Forest classifier. The optimization of hyperparameters was done using grid search and 5-fold stratified cross-validation. To consider the imbalance in outcomes, average precision was taken as the primary scoring metric.
- Threshold Tuning: An inner cross-validation loop was employed to adjust classification thresholds, optimizing balanced accuracy rather than relying on the default 0.5 cutoff.
- Performance Evaluation: Average Precision, ROC-AUC, Balanced Accuracy, and F1-Macro metrics were used to evaluate model performance.

- Feature Importance Assessment: Permutation importance was computed on the held-out test set using the trained Random Forest model to assess each feature's contribution to out-of-sample predictive performance, a practice recommended for evaluating generalization rather than in-sample fit (scikit-learn developers, 2024).
- Visualization and Interpretability: Feature importance plots, and representative decision trees were generated to improve interpretability and transparency of the models.

Robust Extension of Permutation Importance

While permutation importance is effective for ranking predictors, it can be sensitive to feature correlations, potentially inflating or masking true importance (Ewald et al., 2024; Molnar, 2020). Although no strong correlation was observed, but still for robust practice and to validate the stability of our top predictors, we employed a Leave-One-Covariate-Out (LOCO) analysis on the two most important features identified by permutation importance.

The LOCO procedure involved:

1. Removing the feature of interest from the dataset.
2. Retraining the Random Forest classifier on the reduced dataset.
3. Measuring the drop in predictive performance (using Balanced Accuracy and Average Precision).

A significant performance drop upon removal indicates that the feature contributes unique predictive power beyond correlated variables. This method aligns with the ROAR framework (Hooker et al., 2019), which emphasizes retraining to account for distributional shifts caused by feature removal.

Given computational constraints, we applied LOCO only to the top two predictors, as recommended for high-dimensional behavioral data (Malhotra, 2020). Correlation matrices confirmed moderate inter-feature correlations, supporting the need for this robustness check

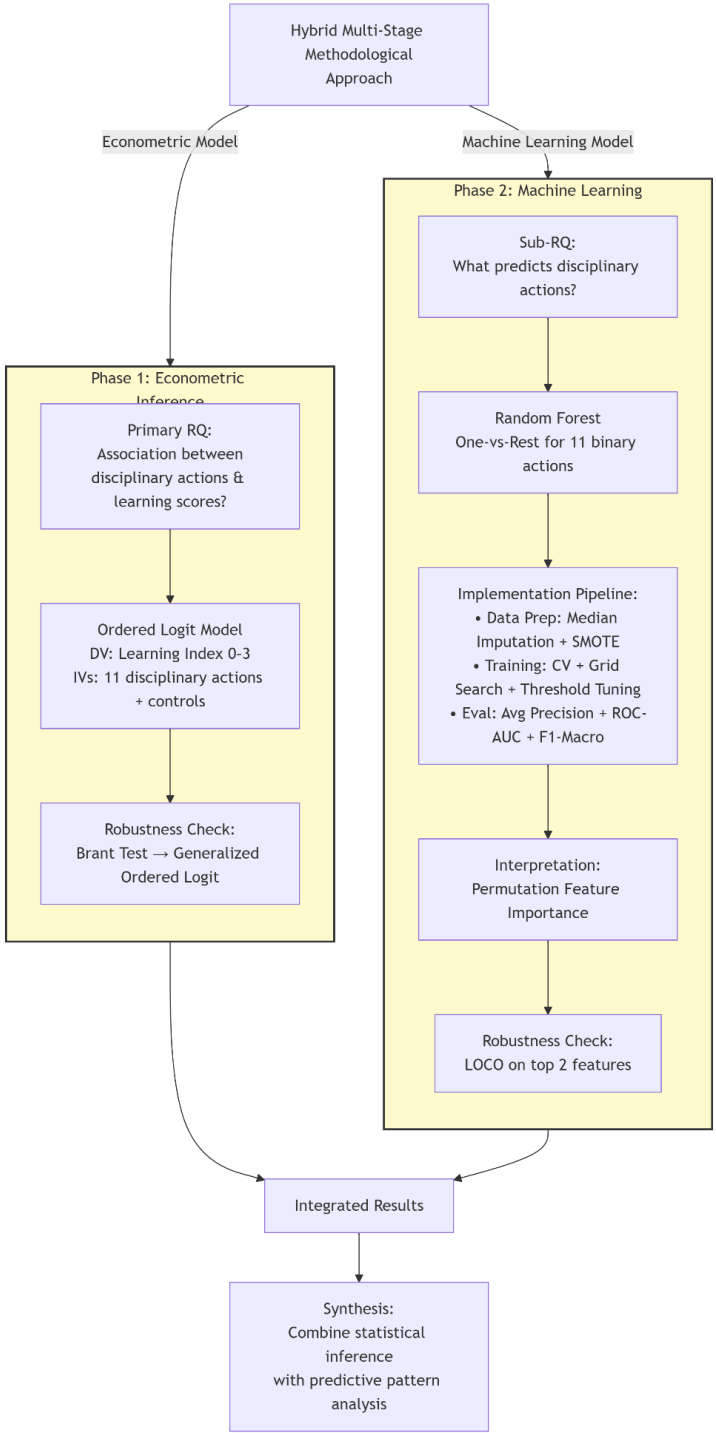


Figure 3: Hybrid multi-stage approach integrating econometric and machine learning model

Results

Marginal Effects on Learning Score Categories

Marginal Effects on Early Learning Outcomes: Socioeconomic Controls and Disciplinary Actions
MICS 2017-2018, Children Aged 3.5-4 Years

Variable	Cat. 0 (No Skills)	Cat. 1 (1 Skill)	Cat. 2 (2 Skills)	Cat. 3 (All Skills)
Socioeconomic Controls				
Wealth index quintile (ref: Richest)				
Poorest	0.1907***	-0.0322***	-0.0490***	-0.1095***
	(0.0153)	(0.0027)	(0.0040)	(0.0090)
Second poorest	0.1168***	-0.0197***	-0.0300***	-0.0671***
	(0.0137)	(0.0024)	(0.0036)	(0.0079)
Middle	0.0911***	-0.0154***	-0.0234***	-0.0523***
	(0.0126)	(0.0022)	(0.0033)	(0.0072)
Fourth richest	0.0468***	-0.0079***	-0.0120***	-0.0269***
	(0.0117)	(0.0020)	(0.0030)	(0.0067)
Richest (ref)	Reference group			
Rural (ref: Urban)	0.0229*	-0.0039*	-0.0059*	-0.0132*
	(0.0095)	(0.0016)	(0.0024)	(0.0055)
Urban (ref)	Reference group			
Mother's Education (ref: None/Preschool)				
Higher	-0.1190***	0.0201***	0.0306***	0.0684***
	(0.0126)	(0.0022)	(0.0033)	(0.0073)
Secondary	-0.0899***	0.0152***	0.0231***	0.0516***
	(0.0114)	(0.0020)	(0.0030)	(0.0066)
Middle	-0.0578***	0.0098***	0.0149***	0.0332***
	(0.0123)	(0.0021)	(0.0032)	(0.0071)
Primary	-0.0358***	0.0060***	0.0092***	0.0206***
	(0.0097)	(0.0016)	(0.0025)	(0.0056)
None (ref)	Reference group			
Female (ref: Male)	-0.0158*	0.0027*	0.0041*	0.0091*
	(0.0064)	(0.0011)	(0.0016)	(0.0037)
Male (ref)	Reference group			
Age (months)	-0.0104***	0.0017***	0.0027***	0.0059***
	(0.0005)	(0.0001)	(0.0001)	(0.0003)
Attended preschool	-0.2481***	0.0419***	0.0638***	0.1425***
	(0.0067)	(0.0016)	(0.0022)	(0.0045)
Functional difficulties (age 2-4 years)	0.0280*	-0.0047*	-0.0072*	-0.0161*
	(0.0152)	(0.0026)	(0.0039)	(0.0087)
Disciplinary Actions				
Took away privileges	-0.0197*	0.0033*	0.0051*	0.0113*
	(0.0078)	(0.0013)	(0.0020)	(0.0045)
Explained behavior	-0.0390***	0.0066***	0.0100***	0.0224***
	(0.0089)	(0.0015)	(0.0023)	(0.0051)

Table 2 – *Continued from previous page*

Variable	Cat. 0 (No Skills)	Cat. 1 (1 Skill)	Cat. 2 (2 Skills)	Cat. 3 (All Skills)
Shook child	0.0026	-0.0004	-0.0007	-0.0015
	(0.0085)	(0.0014)	(0.0022)	(0.0049)
Shouted/yelled at child	0.0321***	-0.0054***	-0.0082***	-0.0184***
	(0.0095)	(0.0016)	(0.0024)	(0.0055)
Gave something else to do	-0.0504***	0.0085***	0.0130***	0.0289***
	(0.0078)	(0.0013)	(0.0020)	(0.0045)
Spanked/hit/slapped	0.0148*	-0.0025*	-0.0038*	-0.0085*
	(0.0079)	(0.0013)	(0.0020)	(0.0046)
Hit with object	0.0185	-0.0031	-0.0048	-0.0106
	(0.0144)	(0.0024)	(0.0037)	(0.0083)
Called dumb/lazy	0.0337***	-0.0057***	-0.0087***	-0.0194***
	(0.0084)	(0.0014)	(0.0022)	(0.0048)
Hit face/head/ears	0.0025	-0.0004	-0.0006	-0.0014
	(0.0083)	(0.0014)	(0.0021)	(0.0047)
Hit hand/arm/leg	-0.0132	0.0022	0.0034	0.0076
	(0.0085)	(0.0014)	(0.0022)	(0.0049)
Beat child hard	-0.0481**	0.0081**	0.0124**	0.0276**
	(0.0154)	(0.0026)	(0.0040)	(0.0088)
Observations	15,590			

Note: Table presents average marginal effects from ordered logit model. Robust standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Reference categories: Richest wealth quintile, Urban residence, Male, No education/preschool. Data source: MICS 2017-2018, children aged 3.5-4 years.

Table 2: presents the average marginal effects from the ordered logit model, which estimates the association between household disciplinary practices and early childhood learning outcomes across four categories, ranging from having no skills to all skills, while controlling for child, parental, and household socioeconomic characteristics.

Disciplinary Actions and Learning Outcomes

From the results (See Table 2) , clear differences emerge across different types of disciplinary actions. Non-violent disciplinary practices are consistently associated with improved learning outcomes. For example, explaining the child's behavior is associated with a 3.9 percentage point reduction in the probability of falling into the lowest learning category (Cat. 0, $p < 0.01$) and a 2.2 percentage point increase in the likelihood of achieving all learning skills (Cat. 3, $p < 0.01$). Similarly, giving the child something else to do shows a strong positive association with learning, reducing the probability of having no skills by 5.0 percentage points and increasing the probability of attaining all skills by 2.9 percentage points (both $p < 0.01$).

In contrast, psychological aggression that includes harsh and verbal forms of discipline is associated with significantly poorer learning outcomes. For instance, children who were shouted or yelled at exhibit a 3.2 percentage point higher probability of having no skills and a 1.8 percentage point lower probability of achieving all skills (both $p < 0.01$). Similarly, being called 'dumb' or 'lazy' is associated with a 3.4 percentage point increase in the probability of having no skills and a 1.9 percentage point decrease in attaining all skills (both $p < 0.01$).

Physical punishment also predominantly shows a negative association with learning. Spanking, hitting, or slapping increases the likelihood of having no skills by 1.5 percentage points, while it reduces the likelihood of achieving all skills by 0.9 percentage points ($p < 0.1$). However, some extreme actions, such as beating the child hard show statistically significant shifts toward higher skill categories that can be explained by the mechanism of short-term compliance (see the Discussion section). Also, some other physical actions, such as shook child or hitting the child with an object do not show statistically significant associations with learning outcomes when controls are included. Marginal effect plot of the ordered logit model is attached below (See Figure 4) . The raw coefficients of the ordered logit results representing the odd log are shown in Table D1 in Appendix D .

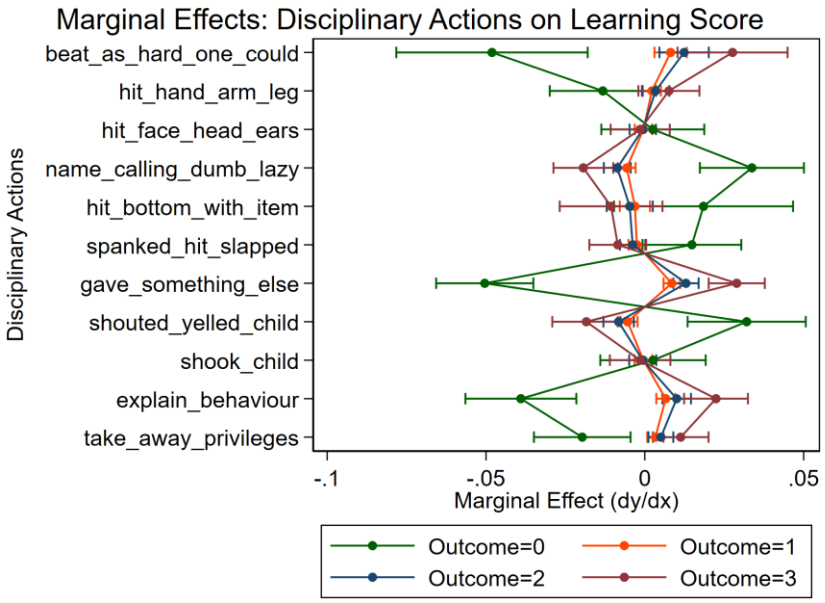


Figure 4: Marginal effect plot of the ordered logit model

Socioeconomic Controls

The findings also showcase that socioeconomic factors are strongly related to learning outcomes. Children from poorer wealth quintiles are significantly more likely to fall into lower learning categories and less likely to achieve all skills compared to children from the richest households ($p < 0.01$ across all categories). Higher levels of maternal education are strongly associated with better learning outcomes, while attending preschool shows the largest positive marginal effect, increasing the probability of attaining all skills by 14.2 percentage points ($p < 0.01$).

Overall, the results showcase that disciplinary actions have a significant impact on early learning outcomes, as non-violent disciplinary actions are associated with an improvement in learning outcomes, while verbal aggression and physical punishment are linked to significantly poorer outcomes. Thus, these findings suggest that not only is the presence of disciplinary action important, but also its nature, in shaping early childhood learning trajectories.

Robust Model Extension Results

Generalized Ordered Logit Model

The Ordered Logit model assumes proportional odds, implying that the effect of each covariate is constant across all thresholds of the Learning Index. Given the skewed outcome distribution, this assumption was tested using the Brant test, which indicated violations for wealth index indicators (windex5, poor, second, fourth), rural residence (rural), child age (cage), school enrollment (school), and selected disciplinary practices, including spanked_hit_slapped, name_calling_dumb_lazy, and hit_face_head_ears.

To address these violations, a Generalized Ordered Logit model was estimated as a robustness check. While coefficient magnitudes differed across thresholds for the variables violating the proportional odds assumption, the direction and statistical significance of effects remained largely consistent with the baseline Ordered Logit results. This indicates that the core findings are not driven by proportional odds violations. Due to the reduced interpretability of threshold-specific coefficients, the standard Ordered Logit model is retained as the primary specification, with the Generalized Ordered Logit results reported in the Appendix D (See Table

D2) for robustness. Visual inspection of the marginal effects plots (See Figure D1) indicates that the Generalized Ordered Logit model produces patterns that are highly consistent with those from the baseline Ordered Logit model, with similar signs, relative magnitudes, and confidence intervals across outcome categories. These figures are presented in the Appendix.

Random Forest Results

To complement the regression analysis, Random Forest classifiers were used to estimate the predictability of different disciplinary practices. This approach was implemented for exploratory purposes and does not allow for causal inference but instead it evaluates the extent to which disciplinary behaviours can be predicted using observed household, caretaker, and child characteristics. The model performance metrics for all disciplinary actions are reported in the Appendix. Given that the disciplinary action responses were highly imbalanced, the accuracy levels from the Random Forest model is acceptable for social science research, with model evaluation focusing on metrics such as average precision and balanced accuracy because of the imbalanced nature of some of the disciplinary actions.

The results of the performance metrics from the models are acceptable for a behavioral data, given the inherent noise in self-reported behavioral data. The model predicts three disciplinary practices moderately well, showing that these behaviors follow clear patterns as these were fairly balanced outcomes. Much of this data includes imbalanced outcomes stemming from the sensitive nature of the outcome variable, influenced by the fact that the data collected was self-reported, which makes Random Forest a good model to predict certain disciplinary actions. Parents explaining why a behavior is wrong is predictable with a good average precision of 0.83 and a balanced accuracy of 0.65. Shouting or yelling, which was the most common disciplinary action, also follows a strong and systematic pattern with an average precision of 0.87 and a balanced accuracy of 0.64. This is followed by shaking the child with an average precision of 0.791 and balanced accuracy of 0.637. Similarly, hitting a child on their ears, face or head is captured with an average precision of 0.69 and a balanced accuracy of 0.64, meaning that the model captures its patterns at acceptable level. In each case, the model distinguishes households that do and do not use these methods better than chance with ROC-AUC values between 0.69 and 0.71, indicating a 69-71% probability that a randomly selected

household using these three disciplinary actions would receive a higher predicted probability than a randomly selected non-using household. Thus, the model distinguishes between adopting and non-adopting households better than chance, suggesting that the use of these disciplinary actions is systematically associated with household-level characteristics captured by the model rather than being purely random (See the performance metric summary table E1 in Appendix E).

Other disciplinary actions display more moderate predictability due to imbalanced outcomes. For instance, taking away a child's privileges has an average precision of 0.58, suggesting greater variation in how and when this practice is used across families. The lower performance metrics for these actions imply that they are less uniformly associated with household characteristics and may be more context-dependent. The disciplinary action of beating the child as hard as one could has a low average precision of 0.087 due to a very high imbalance in the data, and the model does not have high predictive power for this variable. Hence only the disciplinary actions with moderate to high performance models and fairly balanced data were examined for the results of permutation . Specifically, hitting as hard as one could and hitting bottom with items, were excluded. Although the permutation results of these disciplinary actions are still reported in Figure 5 they were not part of our analysis.

The permutation results reveal strong patterns within the dataset. Overall, the caretaker's belief in whether physical punishment is necessary emerges as the strongest and most consistent predictor across disciplinary actions, including both violent and non-violent disciplinary actions (See Figure 5 and Figure E2 in Appendix E). It is especially high for physical disciplinary measures such as spanking, hitting, slapping, and hitting the face, head, or ears, but it is also strongly associated with non-violent actions such as explaining to the child why their behaviour was wrong. This consistency across multiple outcomes suggests that disciplinary practices are highly associated with underlying parental attitudes as well as the child's household environment rather than being isolated responses.

Perceived child aggression compared to peers is the second most important predictor across disciplinary practices, though this is less defined and powerful compared to belief in physical violence being necessary. This variable is especially influential for harsh verbal and physical punishment, with children perceived as more aggressive being more likely to experience

shouting, being called names, or physical discipline. It also predicts non-violent forms of discipline, indicating that perceptions of child behaviour predicts both whether discipline occurs and the type of discipline used. Similarly, whether the child has access to age-appropriate toys emerges as the third most common predictor. Households with toys are more closely associated with non-violent disciplinary strategies, such as redirecting the child's attention or taking away privileges.

The LOCO robustness analysis validates the critical importance of both top predictors identified through permutation importance. Removing "Belief that Physical Discipline is Necessary" resulted in consistent and substantial performance degradation across disciplinary outcomes, with average drops of 3.25% in balanced accuracy, 4.47% in ROC-AUC, and 4.73% in average precision. These declines are particularly significant given the models' moderate baseline performance (balanced accuracy ranging from 0.60 to 0.65), indicating that this caregiver belief provides unique discriminative information that cannot be compensated for by other predictors in the model. Similarly, removing "Aggressive Behavior Compared to Peers" led to meaningful drops in predictive performance, averaging 1.09% in balanced accuracy, 2.04% in ROC-AUC, and 1.70% in average precision. While smaller in magnitude compared to the belief variable, these decreases remain substantial when contextualized against the baseline performance levels and demonstrate the independent contribution of this child behavioral indicator. Following the guidance of Ewald et al. (2024), these robustness results confirm that both features contribute unique predictive power beyond correlated variables, as evidenced by significant performance drops upon removal - a pattern that substantiates the permutation importance rankings even in the presence of moderate inter-feature correlations. The LOCO procedure aligns with the recommended framework for validating feature importance in the presence of dependencies, providing confidence that these associations reflect genuine feature-target relationships rather than artifacts of the correlation structure (See Table F1 and F2 in Appendix F).

Overall, this exploratory analysis highlights that disciplinary practices can be predicted from observed characteristics, with caretaker beliefs playing an important role (See a single decision tree from the random forest model in Figure E3 in Appendix E) .

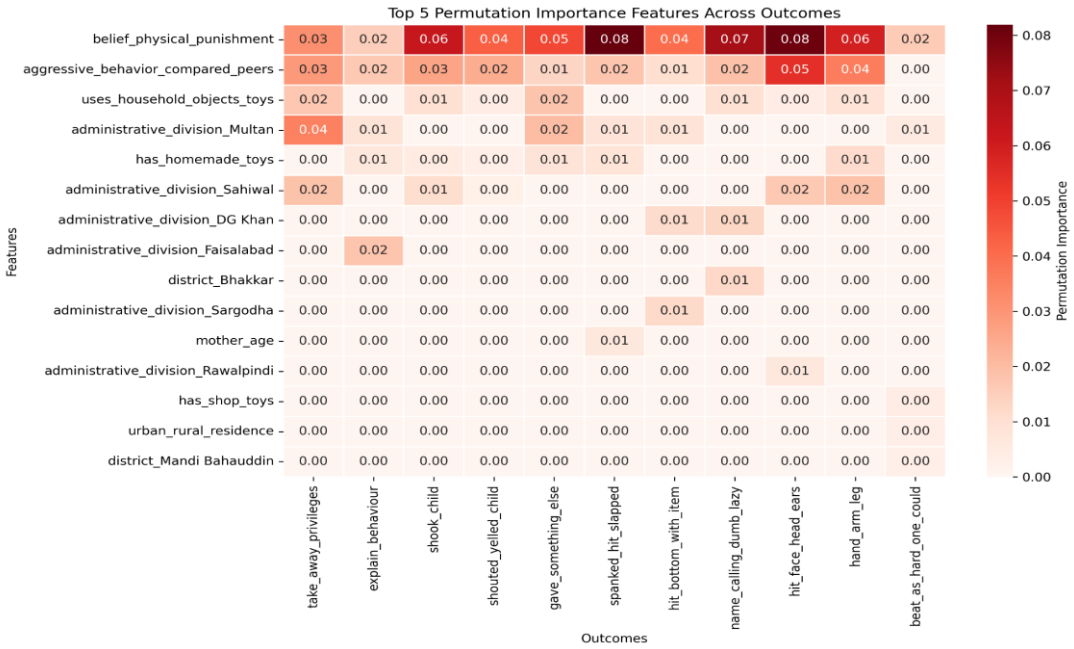


Figure 5: Shows a heat map of the most common predictors of each disciplinary practice included in the Random Forest model. A higher number and a darker colour indicates that the feature is a more prominent predictor of the outcome

Discussion

Overall, the findings show that different forms of household discipline are associated with early learning trajectories. A clear and consistent positive association between non-violent discipline and learning outcomes suggests that these practices function as learning opportunities rather than just ways to discipline children.When caregivers explain a child's behavior or redirect their attention, they appear to engage children both cognitively and emotionally, integrating discipline into the learning process itself.. This is also evident in Piaget (1952) and Vygotsky's (1978) constructivist approaches which explain that learning happens through social interaction, thus adults serve as "critical scaffolds for cognitive development" of children (p.22). Hence, such non-violent disciplinary action encourages them to reflect on and even understand the consequences of their actions. This then enhances their cognitive development and emotional regulation, which are both very important for learning (Piaget, 1952; Vygotsky, 1978).

In contrast, the negative relationship between learning and psychological aggression consistency across all the learning categories in the results of the regression suggests that this has

a broad and pervasive impact on early development. This then links to research that shows that parental verbal abuse alters the integrity of key neural pathways, including those involved in language processing, which can result in deficits in verbal IQ and comprehension (Choi et al., 2009). This detrimental impact of fear-based discipline is then exhibited by how fear displaces a child's focus from learning to avoiding punishment, which then further even negatively impacts their cognitive development.

Similarly, some forms of physical punishment and learning outcomes have a negative association in the regression results, which is also evident from psychological frameworks that state that physical punishment is associated with mental health issues, reduced self-esteem, and increased aggression (Gershoff & Font, 2016). Thus, these outcomes negatively impact children's engagement and limit their educational development in ways that are consistent with the patterns in this study. Also the lack of statistical significance for some physical punishment variables in the study can also be explained by the limited variation and severe imbalance in their reporting within the sample. For instance, only a minority of caregivers, about 6% reported hitting a child with an object as a form of punishment whereas nearly all caregivers within the sample reported that they did not use such methods thus limiting the extent to which the model could detect robust associations.

However, the positive association between beating a child hard and learning outcomes is different from the overall pattern observed in this study between physical discipline and early learning. This may then reflect the disciplinary action paradox, as highlighted in the literature review where corporal punishment is often justified on the basis of immediate obedience and improved classroom behavior. This is evident as the fear of severe punishment leads children to stop misbehaving immediately, which improves outcomes such as attentiveness, task completion, and performance, enhancing their learning (Gershoff, 2002). However, this effect is misleading, as the behavior change is only temporarily driven by fear, rather than the development of skills (Gershoff, 2002). Thus, in older age groups negative consequences of such discipline become evident as children aged 7 to 14 years from the same MICS dataset exposed to severe physical punishment show significantly higher learning difficulties, problems with memory and concentration, anxiety, depression, and impaired social functioning (Kamran & Kazi, 2024).The Random Forest results also show poor predictability for severe physical punishments, such as

beating the child as hard as one could. These outcomes show low precision and weak overall performance of the model due to imbalanced outcomes. This suggests that severe violence is not a regular parenting strategy, but instead can occur in the spur of the moment due to stress or loss of control. Outcomes like this become harder to capture using standard household characteristics and may also be underreported as parents may not want to admit to these forms of discipline in fear of backlash since the data collected is self reported. This also helps explain why severe physical punishment shows weaker and less consistent associations with learning outcomes in the regression analysis.

One of the most powerful results from the machine learning model, the belief about physical punishment being necessary, is the most important predictor for almost all disciplinary actions and supports psychological theories of the attitudinal cycle of violence. These theories suggest that disciplinary behavior is learned and transmitted through beliefs about what is considered acceptable parenting. Parents who believe that physical punishment is necessary are more likely to use harsh verbal and physical discipline and less likely to use non-violent constructive methods. These beliefs shape not only individual actions but the entire approach to childcare, resulting in harsher discipline, which further impacts learning outcomes in children. This means that a child's learning scores have an association with their parents' beliefs on the necessity of physical discipline.

Another takeaway from the exploratory analysis is that many disciplinary practices are not random behaviors but instead, part of stable parenting patterns and the household environment. This is important because such repeated behaviors together shape the child's everyday learning environment. According to labelling theory, a child that grows up hearing negative phrases about themselves tends to internalise them while positive language acts as a protective factor, showing how the child's environment has a very important role in their growth (Lungu, 2016). Repeated behaviors shape the child's everyday learning environment, and how a parent disciplines their child is not an isolated factor, but something that actively influences their development and learning abilities.

Furthermore, one more key insight is that perceived child aggression is an important predictor of how a child is disciplined. Children who are perceived by their parents as more

aggressive than their peers are more likely to experience harsh verbal and physical discipline. This finding brings forward the existence of a discipline-behavior feedback loop where difficult child behavior increases the likelihood of harsh discipline. However, at the same time, harsh discipline may worsen emotional regulation and behavior in young children, reinforcing the cycle and making them more aggressive. Aggressive behaviour in the child and the disciplinary actions they face may be endogenous to each other, as they may influence each other directly. Bandura et al.'s (1961) experiment on the Social Learning Theory shows how children are more likely to imitate aggression, meaning that they are more aggressive if they see a role model act aggressively, and less aggressive if the role model does not act with aggression, meaning that a child's aggressive behaviour can be influenced by the disciplinary action they are subjected to, while the Random Forest model finds perception that the child is aggressive to be a predictor of the disciplinary action. Although the Random Forest analysis cannot determine causality, the child's aggression being an important predictor for some disciplinary actions indicates how discipline and behavior are linked together. Child aggression also predicts the use of non-violent discipline in some households, which may imply that parental beliefs and home environment shapes how behavioral challenges are handled.

Finally, for children under five, early experiences shape the brain in ways that have lasting effects on future learning and productivity, meaning that how they are disciplined at home plays an important role in future human capital formation, tying back to Doyle's (2025) argument on how the early years of a child's life are important in improving human capital. The results in this paper show that certain harsher forms of discipline are associated with lower early learning outcomes, while more constructive practices are linked to better performance. Given that disciplinary behaviours follow predictable patterns within households, when harsh discipline becomes common, its negative effects on learning are more likely to accumulate, leaving children less prepared at school and reinforcing cycles of underachievement that constrain broader human capital development. The World Bank (2024) also states how investments in early childhood yield some of the highest returns in development, and this paper finds how certain disciplinary practices can work directly against these investments. While changing parental behavior is challenging, the fact that these practices are patterned and predictable implies that they can still be molded.

Limitations

Although this study's integration of an econometric model with machine learning represents a methodological contribution and provides a comprehensive view of both associations and predictors of disciplinary practices, it does face certain limitations. Firstly, the analysis establishes an association rather than a causal relationship between parental beliefs, disciplinary practices, and learning outcomes. Secondly, disciplinary practices and learning outcomes are based on the caregiver self-reports in the MICs dataset due to which they may be affected by recall bias or social desirability bias particularly for severe forms of punishment. Thirdly, despite robustness checks, the distribution of learning outcomes is skewed towards lower categories, which may affect precision at higher thresholds. Finally, the possibility of reverse causality cannot be ruled out as children with poorer learning outcomes or more challenging behaviors may face harsher disciplinary responses from caregivers, rather than disciplinary practices solely shaping learning trajectories. In the Pakistani context, where disciplinary norms are deeply embedded in cultural and religious traditions (Ashraf & Holden, 2022), caregiver beliefs may be particularly resistant to change in response to individual child outcomes. While we cannot definitively establish causal direction, several factors suggest that the relationships depicted in our analysis are meaningful. First, beliefs about discipline - especially those rooted in cultural norms and shaped by socioeconomic factors -tend to form early and exhibit considerable stability (Hirsjarvi & Perälä-Littunen, 2001; List et al., 2021). Hirsjarvi and Perälä-Littunen (2001) document that parental beliefs demonstrate "resistance to change…even in the face of a great deal of rational evidence" (p. 89) and are transmitted across generations through cultural and institutional sources. Complementing this, List et al. (2021) show clear socioeconomic gradients in parental beliefs that emerge very early in life, with these beliefs subsequently predicting child outcomes measured months and even years later. Their longitudinal analysis demonstrates that parental beliefs measured at 13-16 months predict language outcomes at 37-40 months, with beliefs alone explaining up to 18% of variation in child language skills-a temporal ordering that reduces concerns about reverse causality.

Second, the theoretical framework underlying our analysis positions beliefs as antecedents to behavior. Developmental psychology research consistently shows that parental beliefs about child development predict parenting practices and child outcomes (Benasich & Brooks-Gunn,

1996; Miller, 1988), functioning as relatively stable psychological constructs that guide behavioral choices. For instance, Kochanska (1990, cited in Hirsjarvi & Perälä-Littunen, 2001) found that maternal beliefs measured at one point had predictive value for maternal control behavior 2-3 years later, suggesting that beliefs precede and shape disciplinary actions rather than merely reflecting responses to child behavior.

Third, experimental evidence further supports the causal influence of parental beliefs on behavior. List et al. (2021) demonstrate through two randomized controlled trials that altering parental beliefs about child development leads to subsequent changes in parenting practices and child outcomes. These experiments provide direct evidence for the belief → behavior pathway that our observational analysis assumes.

Nevertheless, we acknowledge that beliefs and practices likely operate in a bidirectional relationship over time: harsh discipline may reinforce authoritarian beliefs, while challenging child behaviors may strengthen caregivers' reliance on punitive strategies (Mills & Rubin, 1992). Without longitudinal data or exogenous variation in disciplinary practices, we cannot fully rule out reverse causality or disentangle cumulative feedback effects. Future research using panel data or natural experiments would strengthen causal claims regarding how beliefs shape disciplinary practices and child learning trajectories in low- and middle-income contexts.

To provide a visual summary of this framework, we depict the relationships between caregiver beliefs, disciplinary practices, and child outcomes in the graph attached in the Appendix G (see Figure G). This framework can also serve as a foundation for formulating hypotheses for further research.

Future Research Direction

While this study establishes a strong association between disciplinary practices and learning outcomes, the potential for endogeneity due to reverse causality remains a limitation. Because the MICS data are cross-sectional and lack longitudinal follow-up, traditional panel methods are not feasible. Future research could nevertheless employ an instrumental variables (IV) approach to better establish a causal link. Specifically, researchers could use the leave-one-out average of

the belief that physical discipline is necessary, aggregated at the cluster or neighborhood level, as an instrument. This measure captures the social norms that shape a caregiver's disciplinary behavior, and the paper's machine-learning results confirm that such beliefs are the strongest predictor of disciplinary practices, while remaining plausibly exogenous to the learning outcomes of any individual child.

Policy

The findings of this study have important implications for early childhood policy, parenting interventions, and human capital development strategies. Together, the econometric and machine learning results show how disciplinary practices follow stable patterns shaped by the environment of the child. This predictability makes disciplinary practices in early childhood a very important target for policy interventions.

One major finding from this paper is that belief that physical punishment is necessary is the strongest predictor of disciplinary behavior across almost all outcomes. This means that policy efforts should focus more on changing the underlying the beliefs parents have about disciplining their children. Parents should be educated on how physical or fear based discipline is not necessary, and instead can be harmful for their children. Moreover, alternatives such as explanation should be promoted as beliefs shape parenting styles and changes in attitudes towards discipline have been associated with a reduction in violent discipline in favour of more non-violent practices which are positively associated with learning outcomes. Policy messaging should also link actions like explaining why their behaviour was wrong to the child directly to child's educational success as parents may be more receptive to change when they understand that how they discipline their child is associated with tangible things like literacy and numeracy skills, rather than viewing discipline purely as a moral or behavioral concern.

The findings also highlight how it is important to integrate parenting practices into broader early human capital investment frameworks. While policies like expanding access to school are important, these investments may not show high returns if children are raised in environments where harsh discipline is common. The continued exposure of children to harsh, violent discipline can counteract many of the investments made to improve their learning outcomes. As such, policymakers should consider parental behaviour to be a component of early

childhood development strategies, not as an independent consideration, when making targeted investments to reduce the use of harsh disciplinary methods.

Conclusion

This study shows how the methods by which children under 5 years of age are disciplined at home play an important role in their early learning outcomes. The results show that non-violent discipline, such as explaining behavior, are linked to better literacy and numeracy outcomes, while certain forms of harsher discipline are associated with poorer learning outcomes. The findings also show that beliefs about whether physical punishment is necessary strongly influences how children are disciplined, meaning that the child's environment at home in their early years is extremely important. As a result, policies that aim to improve early learning in Pakistan need to go beyond just increasing school access or resources and also address parenting beliefs and practices within the home. Improving early childhood discipline is therefore not only about reducing harm, but also about strengthening the foundations of learning and long-term human capital development. Methodologically, our hybrid approach- offers a more reliable framework for analyzing complex behavioral data. By validating permutation importance through retraining, we mitigate biases from feature correlations, enhancing the credibility and applicability of interpretable machine learning in social science research, where transparency and replicability are paramount.

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Appendix A



Figure A1: This histogram shows the distribution of the wealth index of the sample of the study.



Figure A2: This histogram shows the distribution of the mother's education of the sample of the study.

Appendix B

Wealth Index

In educational research, the most robust predictor is the socio-economic status of the child which has a medium-to-strong relationship with academic achievement. Socio-economic status exhibits a consistent, medium-to-strong relationship with academic achievement (average correlation r = 0.29), establishing it as one of the most robust predictors in educational research (Sirin, 2005).

Mother's Education

Maternal education is said to have a causal impact in the context of Pakistan. This is primarily due to increased time spent on educational activities at home and improving home learning environment. It is therefore a strong confounding variable that must be controlled for when estimating effect on learning outcomes (Andrabi et al., 2009).

Area Type (Urban/Rural)

Literature shows a huge urban-rural divide in learning outcomes for children, with rural students consistently showing lower reading literacy than urban students. These gaps arise from weaker home support, poorer school infrastructure and teaching resources in the rural regions (Zhang, 2006).

Child Gender

The gender gap is also evident in learning outcomes, with females outperforming males in literacy and males achieving higher in numeracy. The pattern is largely attributed to variation in motivation and confidence driven by social and cultural factors, including gender stereotypes and differential encouragement (Pugu & Aslan, 2025).

Child Age

The evidence for the inclusion of child's age comes from large-scale assessment like ECDI, which uses age (e.g., 36-59 months) as the framework for measuring developmental progress, theoretically grounded on the fact that developmental and learning competencies are inherently age-dependent (Loizillon et al., 2017).

Attendance of School Program

A study in Punjab, Pakistan, found that structured Early Childhood Education (ECE) programs significantly boost cognitive development and foundational literacy and numeracy skills (Yasrab & Shah, 2022).

Child Disability Status

Based on research, children with disabilities perform worse in both literacy and numeracy compared to children without disabilities. This is supported by research conducted in rural Punjab, Pakistan, that while children with disabilities are enrolled in mainstream schools, they demonstrate substantially lower basic skills than their non-disabled peers (Malik et al., 2020).

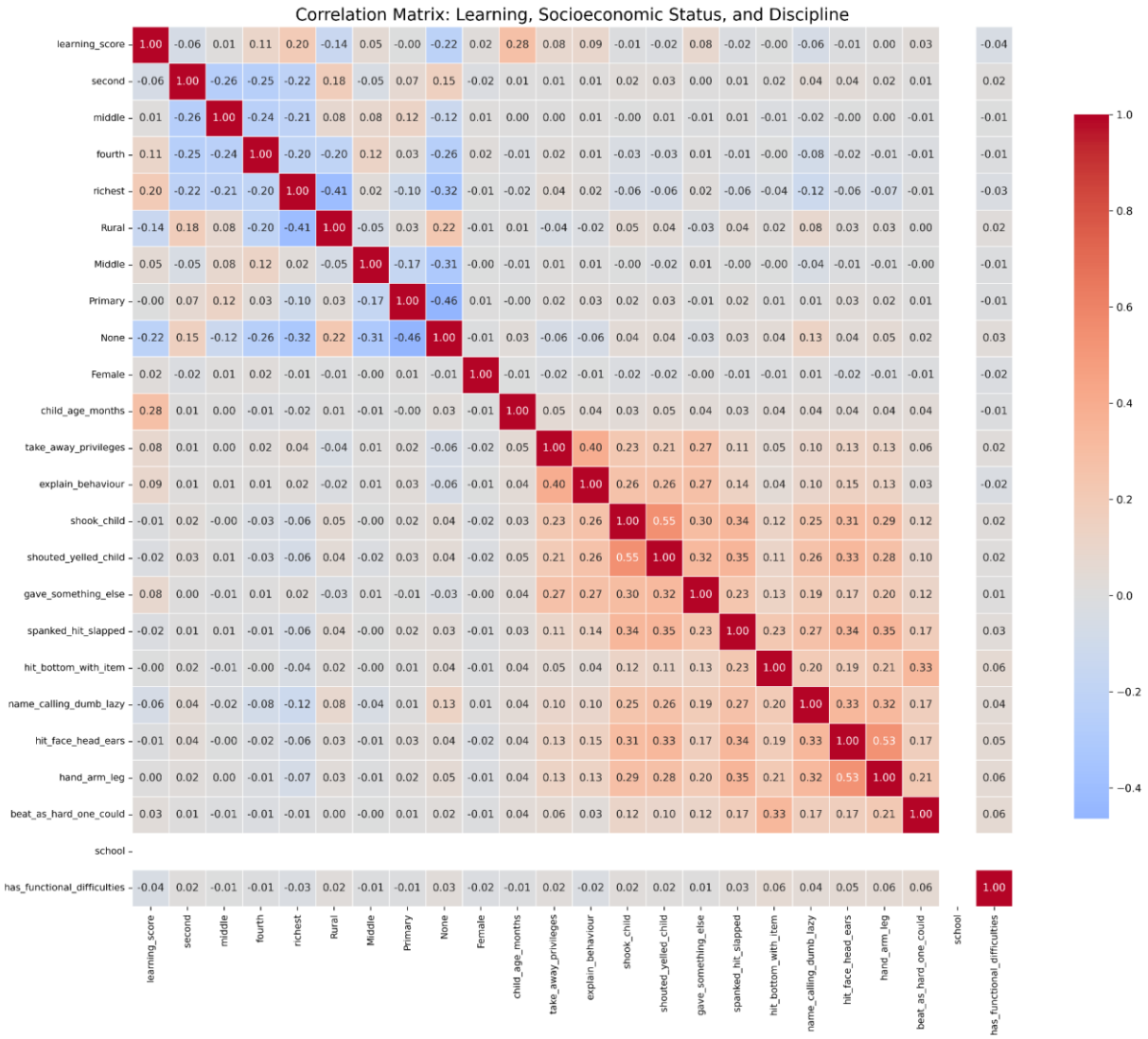


Figure B1: Correlation matrix , showing the pair wise correlation of the covariates in the econometric models

Appendix C

Thematic Group	Key Supporting Citations	Variables in This Group

Child Characteristics & Demographics	Often used as control variables (e.g., Ibrahim & Patwary, 2025; Saeedzai et al., 2025; MacKenzie et al., 2012; Evans, 2015)	child_sex, child_age_months, height_for_age_zscore, weight_for_age_zscore, weight_for_height_zscore, body_mass_index
Child Functional Difficulties & Behavior	Engfer & Schneewind (1982) , "difficult" child as predictor. McElroy & Rodriguez (2008) - children with externalizing behaviors. Ateah & Durrant (2005)- response to misbehavior.	difficulty_seeing, difficulty_hearing, difficulty_walking_compared_peers, difficulty_picking_up_objects, difficulty_understanding_caretaker, difficulty_being_understood, difficulty_learning, difficulty_playing, aggressive_behavior_compared_peers , has_functional_difficulties
Parental Education & Knowledge	Cappa & Khan (2011); Dietz (2000); Ibrahim & Patwary (2025); Islam (2024); Malik (2010); Saeedzai et al. (2025) - lower education predicts harsher discipline.	mother_education_level, household_head_education

Household Socioeconomic Status	Beatriz & Salhi (2019); Choi et al. (2018) - lower wealth predicts violent discipline. Dietz (2000); Evans (2015) - fewer resources linked to harsh punishment. Cappa & Khan (2011) - rural disadvantage.	wealth_quintile, urban_rural_residence, area_type
Caregiver Demographics & Structure	Evans (2015) - teenage parenthood as risk. Rizvi & Najam (2019); Malik (2010) - family size/structure.	mother_age, household_head_sex
Early Childhood Stimulation & Resources	Related to socioeconomic disadvantage (Dietz, 2000; Beatriz & Salhi, 2019) - lack of resources increases risk.	ever_attended_preschool, attended_early_childhood_education, number_childrens_books, has_homemade_toys, has_shop_toys, uses_household_objects_toys
Parental Attitudes & Norms on Violence	Ateah & Durrant (2005); Jackson et al. (1999); Cappa & Khan (2011) - approval of physical punishment. Nho & Seng (2017) - acceptance of spousal abuse.	domestic_violence_perception, belief_physical_punishment

Geographic/Contextual Controls	Beatriz & Salhi (2019); Cappa & Khan (2011) – regional/country-level variation in norms and economics. Malik (2010) – regional differences within Pakistan.	administrative_division_DG Khan, administrative_division_Faisalabad, administrative_division_Gujranwala, administrative_division_Lahore, administrative_division_Multan, administrative_division_Rawalpindi, administrative_division_Sahiwal, administrative_division_Sargodha,

Table C : The theoretical justification of input variables in machine learning models.

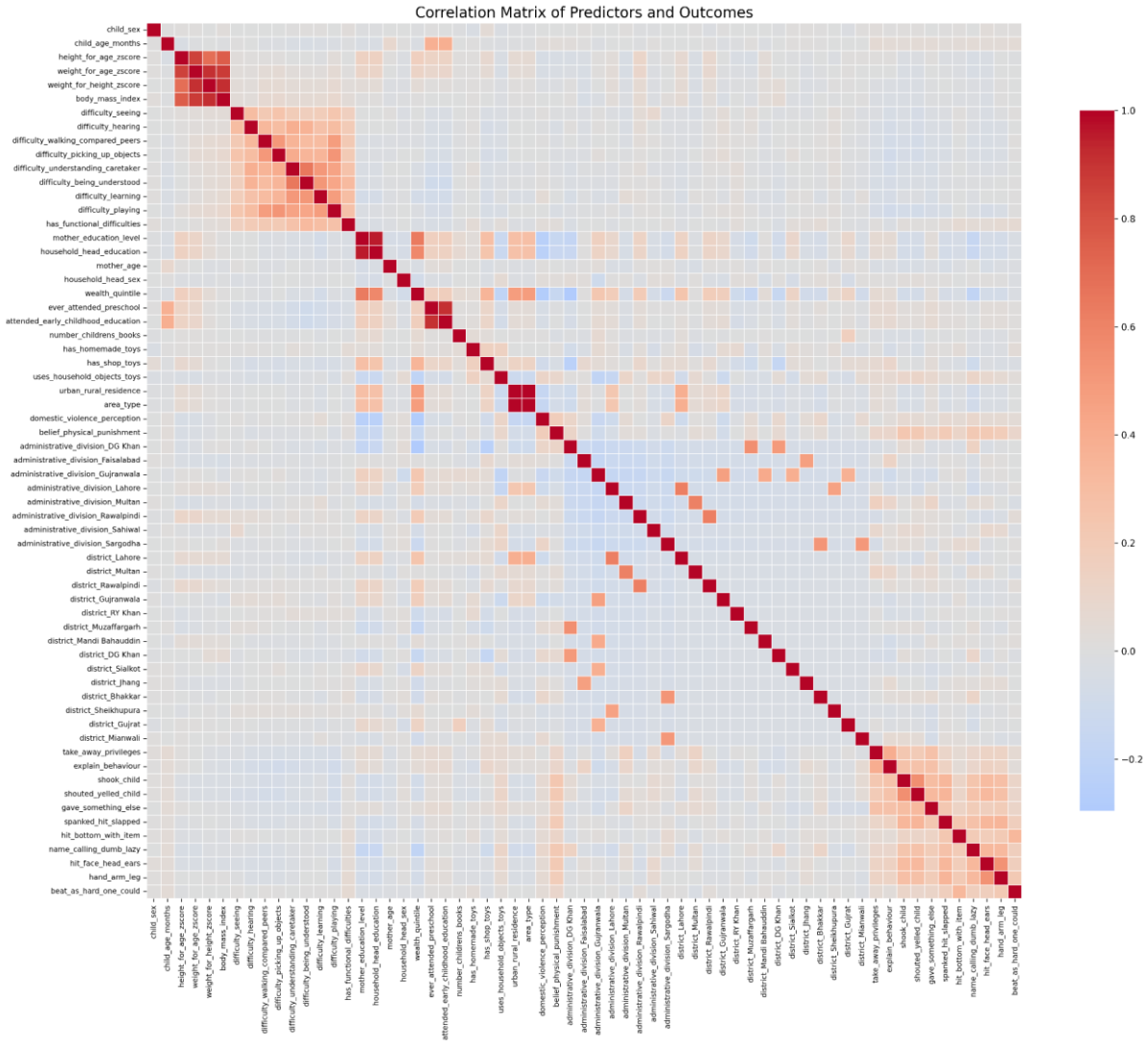


Figure C1: Correlation matrix of the inputs of the random forest inputs

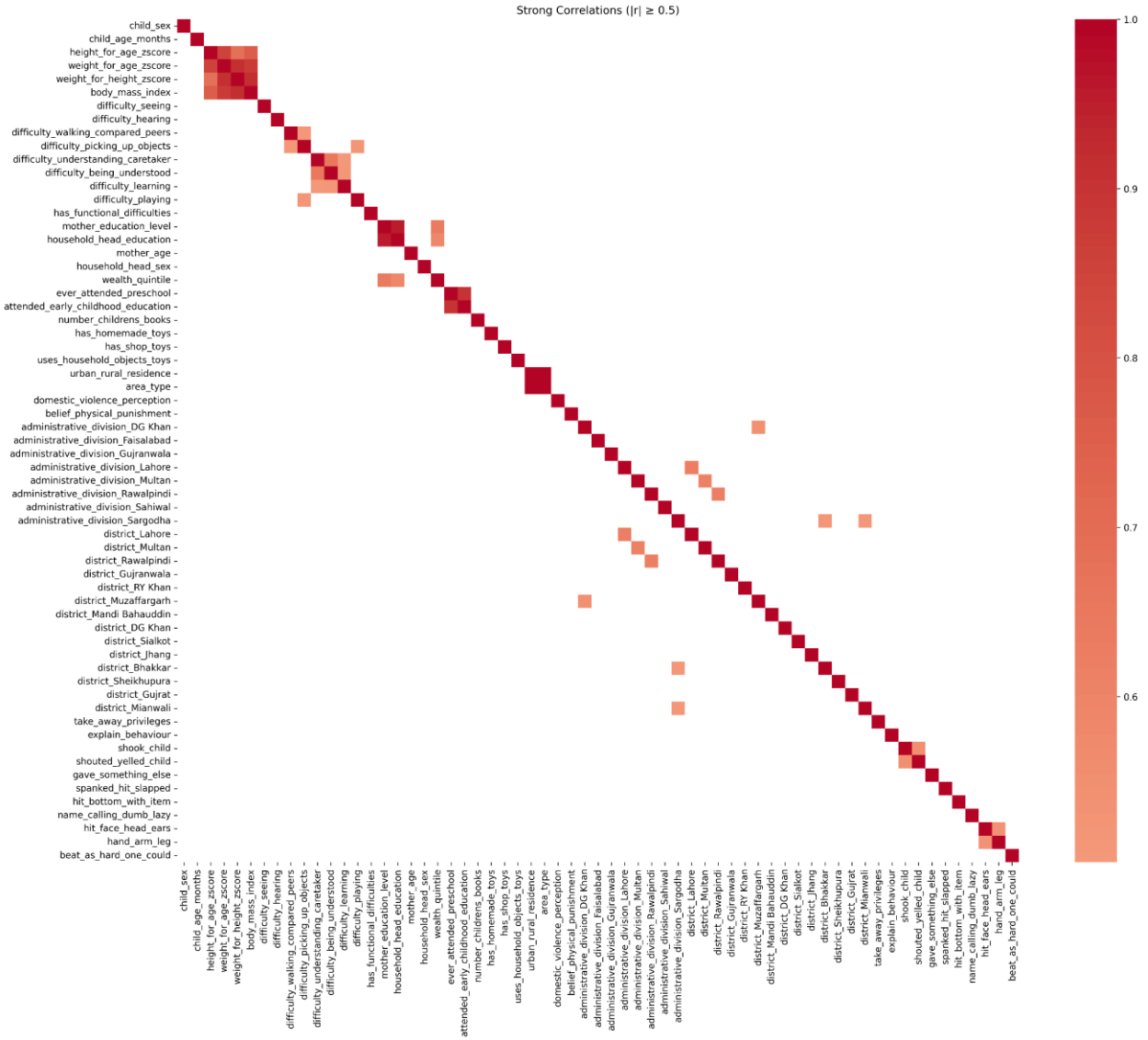


Figure C2: Correlation matrix displaying the pair wise correlations greater than 0.5

Appendix D

Generalized Ordered Logit Coefficients: Learning Score by Wealth Quintile

Generalized Ordered Logit Coefficients: Wealth Quintiles, Disciplinary Actions, and Learning Scores
MICS 2017-2018, Children Aged 3.5-4 Years

Variable	Coefficient (1)
0	
Poorest	-1.072***
	(0.0826)
Second poorest	-0.688***
	(0.0737)
Middle wealth	-0.513***
	(0.0670)
Fourth richest	-0.283***
	(0.0651)
Rural residence	-0.0686
	(0.0542)
Higher education	0.508***
	(0.0616)
Secondary education	0.356***
	(0.0563)
Primary education	0.0992*
	(0.0475)
Female	0.0826*
	(0.0339)
Age (months)	0.0474***
	(0.00292)
Functional difficulties (age 2-4 years)	-0.149
	(0.0812)
Took away privileges	0.106*
	(0.0412)
Explained behavior	0.213***
	(0.0473)
Shook child	-0.0117
	(0.0449)
Shouted/yelled at child	-0.173***
	(0.0503)
Gave something else to do	0.269***
	(0.0416)
Spanked/hit/slapped	-0.0373
	(0.0451)
Hit with object	-0.110
	(0.0764)

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Table 3 – *Continued from previous page*

Variable	Coefficient (1)
Called dumb/lazy	-0.224***
	(0.0478)
Hit face/head/ears	0.0214
	(0.0471)
Hit hand/arm/leg	0.0690
	(0.0453)
Beat child hard	0.257**
	(0.0823)
Attended preschool	1.277***
	(0.0427)
Constant	-2.791***
	(0.155)
1	
Poorest	-1.333***
	(0.0882)
Second poorest	-0.769***
	(0.0765)
Middle wealth	-0.513***
	(0.0670)
Fourth richest	-0.303***
	(0.0671)
Rural residence	-0.116*
	(0.0547)
Higher education	0.508***
	(0.0616)
Secondary education	0.356***
	(0.0563)
Primary education	0.0992*
	(0.0475)
Female	0.0826*
	(0.0339)
Age (months)	0.0659***
	(0.00331)
Functional difficulties (age 2-4 years)	-0.149
	(0.0812)
Took away privileges	0.106*
	(0.0412)
Explained behavior	0.213***
	(0.0473)
Shook child	-0.0117
	(0.0449)
Shouted/yelled at child	-0.173***
	(0.0503)

Continued on next page

Table 3 – *Continued from previous page*

Variable	Coefficient (1)
Gave something else to do	0.269***
	(0.0416)
Spanked/hit/slapped	-0.111*
	(0.0478)
Hit with object	-0.110
	(0.0764)
Called dumb/lazy	-0.160**
	(0.0502)
Hit face/head/ears	-0.00312
	(0.0478)
Hit hand/arm/leg	0.0690
	(0.0453)
Beat child hard	0.257**
	(0.0823)
Attended preschool	1.454***
	(0.0474)
Constant	-4.407***
	(0.173)
2	
Poorest	-1.205***
	(0.106)
Second poorest	-0.598***
	(0.0881)
Middle wealth	-0.513***
	(0.0670)
Fourth richest	-0.144*
	(0.0760)
Rural residence	-0.243***
	(0.0652)
Higher education	0.508***
	(0.0616)
Secondary education	0.356***
	(0.0563)
Primary education	0.0992*
	(0.0475)
Female	0.0826*
	(0.0339)
Age (months)	0.0667***
	(0.00427)
Functional difficulties (age 2-4 years)	-0.149
	(0.0812)
Took away privileges	0.106*
	(0.0412)

Continued on next page

Table 3 – *Continued from previous page*

Variable	Coefficient (1)
Explained behavior	0.213***
	(0.0473)
Shook child	-0.0117
	(0.0449)
Shouted/yelled at child	-0.173***
	(0.0503)
Gave something else to do	0.269***
	(0.0416)
Spanked/hit/slapped	-0.167**
	(0.0577)
Hit with object	-0.110
	(0.0764)
Called dumb/lazy	-0.0753
	(0.0617)
Hit face/head/ears	-0.131*
	(0.0594)
Hit hand/arm/leg	0.0690
	(0.0453)
Beat child hard	0.257**
	(0.0823)
Attended preschool	1.268***
	(0.0593)
Constant	-5.172***
	(0.218)
Observations	15,590

Note: Table presents generalized ordered logit coefficients. Standard errors in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Reference categories: Richest wealth quintile, Urban residence, Male, No education/preschool. Data source: MICS 2017-2018, children aged 3.5-4 years.

Table D1: Raw Coefficients from the Generalised Ordered Logit Model

Generalized Ordered Logit Marginal Effects on Learning Score Categories

Generalized Ordered Logit Marginal Effects on Early Learning Outcomes: Socioeconomic Controls and Disciplinary Actions
MICS 2017-2018, Children Aged 3.5-4 Years

Variable	Cat. 0 (No Skills)	Cat. 1 (1 Skill)	Cat. 2 (2 Skills)	Cat. 3 (All Skills)
Socioeconomic Controls				
Wealth index quintile (ref: Richest)				
Poorest	0.2056***	-0.00775	-0.06963***	-0.1282***
	(0.01538)	(0.00896)	(0.00844)	(0.01140)
Second poorest	0.1320***	-0.01784**	-0.05054***	-0.06362***
	(0.01400)	(0.00796)	(0.00727)	(0.00939)
Middle	0.09830***	-0.02219***	-0.02154***	-0.05457***
	(0.01276)	(0.00295)	(0.00288)	(0.00716)
Fourth richest	0.05421***	-0.00922	-0.02969***	-0.01531*
	(0.01248)	(0.00754)	(0.00646)	(0.00809)
Richest (ref)	Reference group			
Rural (ref: Urban)	0.01316	0.00405	0.00865	-0.02586***
	(0.01038)	(0.00689)	(0.00583)	(0.00690)
Mother's Education (ref: None/Preschool)				
Higher	-0.09742***	0.02199***	0.02135***	0.05409***
	(0.01171)	(0.00272)	(0.00268)	(0.00655)
Secondary	-0.06827***	0.01541***	0.01496***	0.03790***
	(0.01074)	(0.00246)	(0.00241)	(0.00600)
Primary	-0.01902**	0.00429**	0.00417**	0.01056**
	(0.00909)	(0.00205)	(0.00199)	(0.00505)
Female (ref: Male)	-0.01584**	0.00358**	0.00347**	0.00880**
	(0.00649)	(0.00146)	(0.00143)	(0.00361)
Age (months)	-0.00908***	-0.00071*	0.00269***	0.00710***
	(0.00055)	(0.00041)	(0.00040)	(0.00045)
Functional difficulties (age 2-4 years)	0.02855*	-0.00644*	-0.00626*	-0.01585*
	(0.01554)	(0.00352)	(0.00341)	(0.00863)
Attended preschool	-0.24488***	0.02911***	0.08084***	0.13493***
	(0.00710)	(0.00515)	(0.00499)	(0.00611)
Disciplinary Actions				
Took away privileges	-0.02032**	0.00459***	0.00445**	0.01128**
	(0.00790)	(0.00178)	(0.00173)	(0.00440)
Explained behavior	-0.04079***	0.00921***	0.00894***	0.02265***
	(0.00904)	(0.00207)	(0.00199)	(0.00504)
Shook child	0.00224	-0.00051	-0.00049	-0.00124
	(0.00861)	(0.00194)	(0.00189)	(0.00478)
Shouted/yelled at child	0.03322***	-0.00750***	-0.00728***	-0.01844***

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Table 4 – *Continued from previous page*

Variable	Cat. 0 (No Skills)	Cat. 1 (1 Skill)	Cat. 2 (2 Skills)	Cat. 3 (All Skills)
	(0.00968)	(0.00219)	(0.00214)	(0.00538)
Gave something else to do	-0.05149***	0.01162***	0.01128***	0.02859***
	(0.00798)	(0.00184)	(0.00179)	(0.00444)
Spanked/hit/slapped	0.00715	0.00927	0.00134	-0.01776**
	(0.00865)	(0.00605)	(0.00546)	(0.00616)
Hit with object	0.02104	-0.00475	-0.00461	-0.01168
	(0.01464)	(0.00332)	(0.00322)	(0.00812)
Called dumb/lazy	0.04294***	-0.01916**	-0.01576**	-0.00801
	(0.00916)	(0.00658)	(0.00588)	(0.00656)
Hit face/head/ears	-0.00410	0.00456	0.01348**	-0.01394**
	(0.00903)	(0.00620)	(0.00553)	(0.00631)
Hit hand/arm/leg	-0.01323	0.00299	0.00290	0.00734
	(0.00869)	(0.00196)	(0.00191)	(0.00483)
Beat child hard	-0.04924***	0.01111***	0.01079***	0.02734***
	(0.01576)	(0.00358)	(0.00348)	(0.00875)
Observations	15,590			

Note: Table presents average marginal effects from generalized ordered logit model for learning score categories 0–3. Standard errors in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Reference categories: Richest wealth quintile, Urban residence, Male, No education/preschool. Data source: MICS 2017-2018, children aged 3.5-4 years.

Table D2 : Average Marginal Effects Results of Generalized Ordered Logit

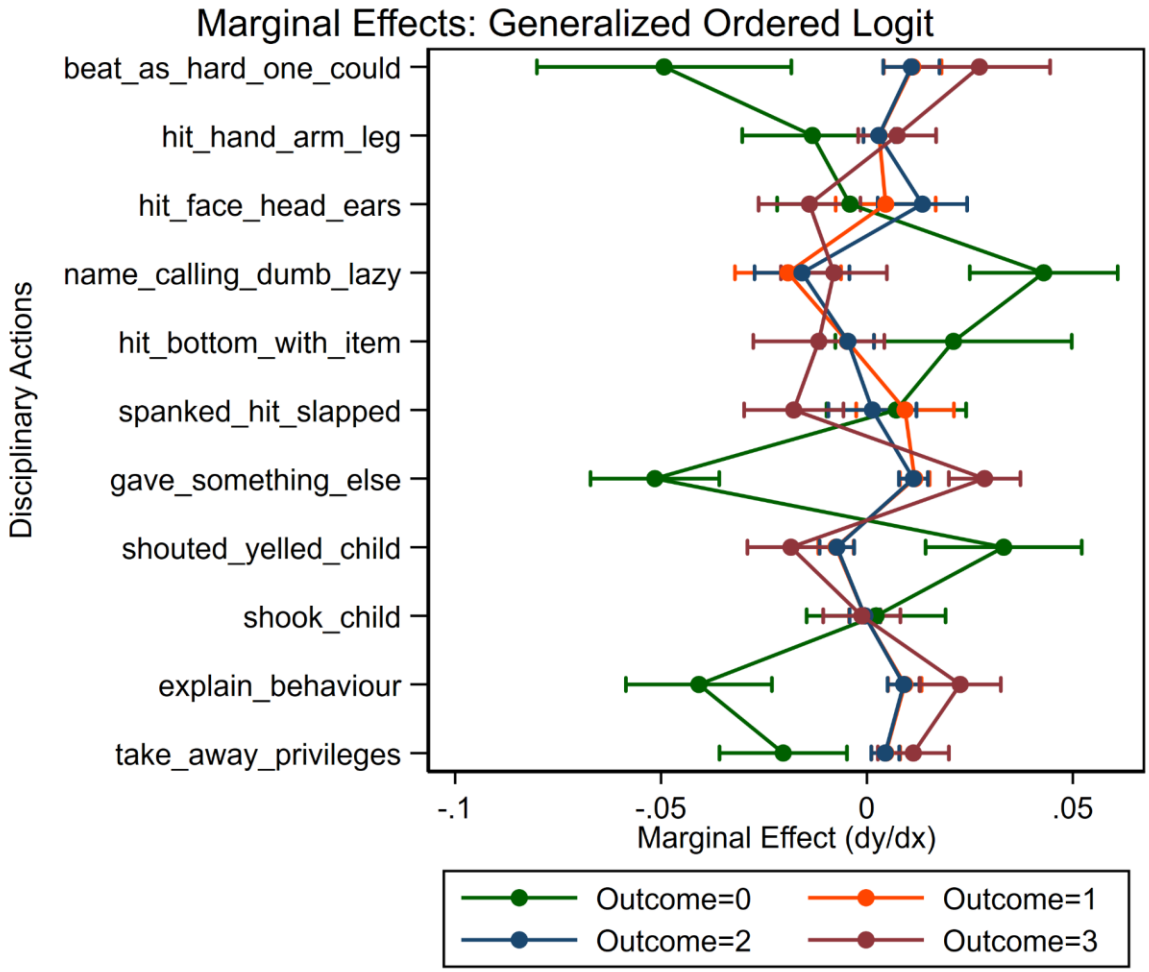


Figure D3: Marginal Effects Plot of the disciplinary actions in Generalized Ordered Logit Model

Appendix E

Disciplinary Action	Average Precision	ROC-AUC	Balanced Accuracy	Test Accuracy	F1-Macro	Optimal Threshold
explain_behaviour	0.831	0.708	0.656	0.643	0.624	0.66
shouted_yelled_child	0.868	0.702	0.644	0.657	0.608	0.65

shook_child	0.791	0.681	0.637	0.632	0.621	0.59
hit_face_head_ears	0.691	0.699	0.648	0.648	0.648	0.50
name_calling_dumb_lazy	0.538	0.705	0.643	0.655	0.628	0.36
take_away_privileges	0.586	0.690	0.636	0.639	0.630	0.41
gave_something_else	0.589	0.660	0.620	0.618	0.616	0.45
spanked_hit_slapped	0.592	0.663	0.610	0.621	0.610	0.48
hand_arm_leg	0.552	0.673	0.622	0.609	0.605	0.39
hit_bottom_with_item	0.178	0.689	0.651	0.688	0.509	0.11
beat_as_hard_one_could	0.086	0.630	0.583	0.722	0.490	0.10

Table E1: Summary of performance metric of the random forest model of the 11 disciplinary actions.

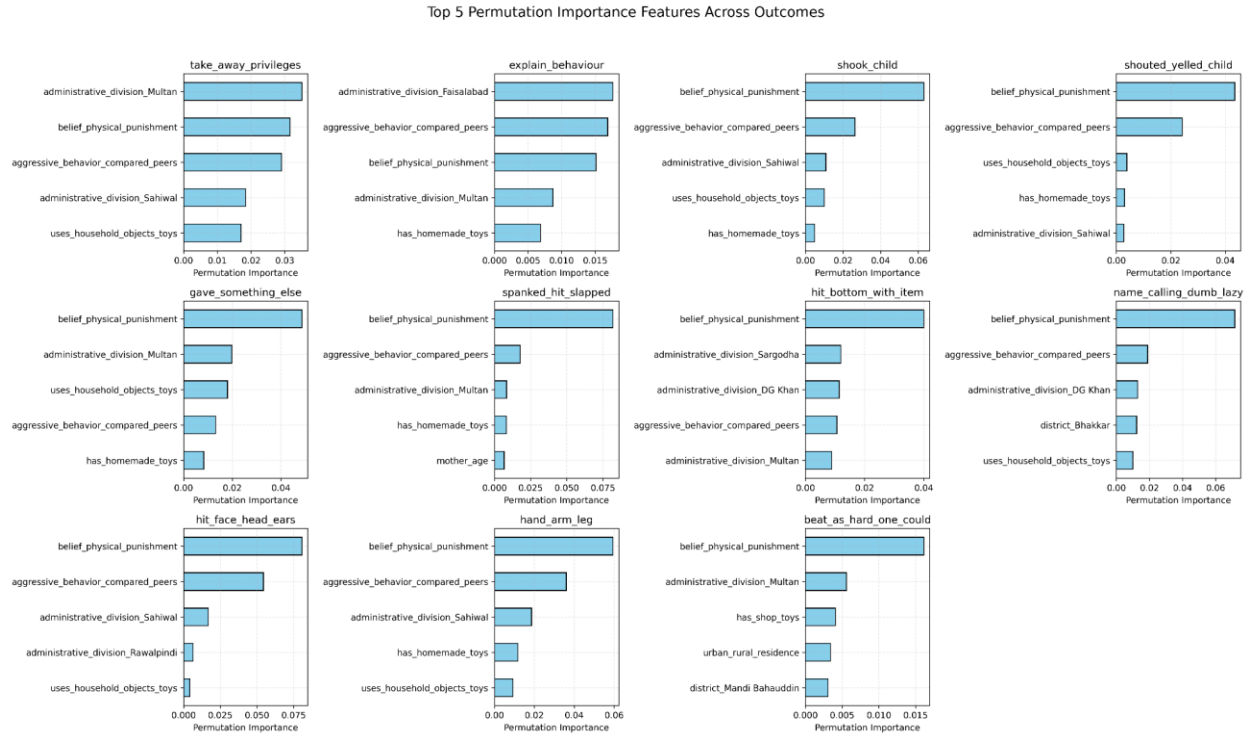


Figure E2: Sub bar plots showing the top 5 permutation feature importance across the 11 disciplinary outcomes

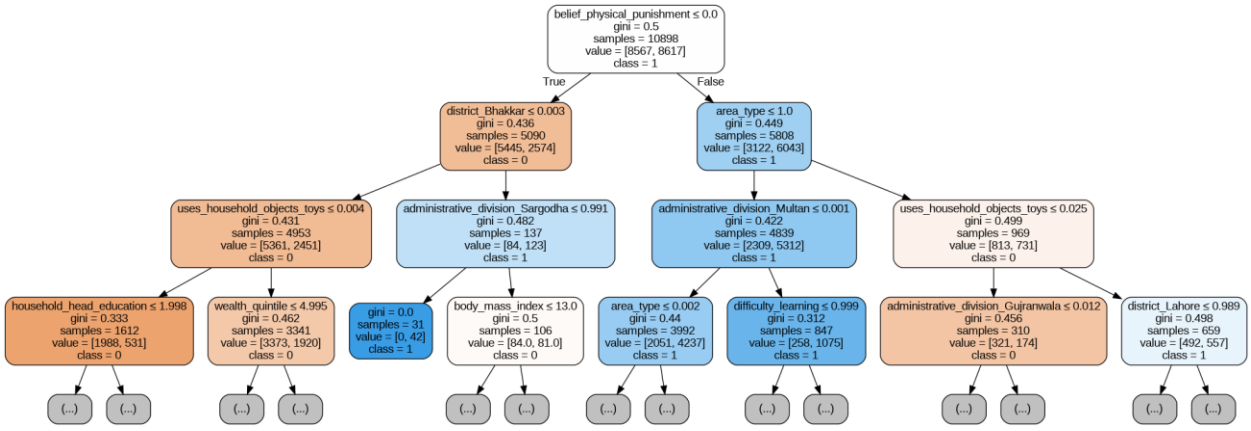


Figure E3: shows a single decision tree from the Random Forest model.

Appendix F

Disciplinary Outcome	Metric	Baseline	LOCO: Without "Belief Physical Discipline Necessary"	LOCO: Without "Aggressive Behavior Compared to Peers"	Drop % (Belief)	Drop % (Aggressive)
take_away_privileges	Balanced Accuracy	0.6335	0.6201	0.6237	-2.12%	-1.55%
	ROC-AUC	0.6853	0.6688	0.6721	-2.41%	-1.93%
	Average Precision	0.5800	0.5636	0.5607	-2.83%	-3.33%
explain_behaviour	Balanced Accuracy	0.6478	0.6379	0.6348	-1.53%	-2.01%
	ROC-AUC	0.7032	0.6872	0.6769	-2.28%	-3.74%

	Average Precision	0.8282	0.8251	0.8136	-0.37%	-1.76%
shook_child	Balanced Accuracy	0.6268	0.6035	0.6199	-3.72%	-1.10%
	ROC-AUC	0.6831	0.6463	0.6711	-5.39%	-1.76%
	Average Precision	0.7959	0.7658	0.7834	-3.78%	-1.57%
shouted_yelled_child	Balanced Accuracy	0.6372	0.6214	0.6201	-2.48%	-2.68%
	ROC-AUC	0.7008	0.6653	0.6727	-5.07%	-4.01%
	Average Precision	0.8674	0.8498	0.8533	-2.03%	-1.63%
gave_something_else	Balanced Accuracy	0.6142	0.5821	0.6051	-5.23%	-1.48%

	ROC-AUC	0.6598	0.6315	0.6502	-4.29%	-1.46%
	Average Precision	0.5857	0.5560	0.5739	-5.07%	-2.01%
spanked_hit_slapped	Balanced Accuracy	0.6160	0.5837	0.6089	-5.24%	-1.15%
	ROC-AUC	0.6601	0.6202	0.6500	-6.05%	-1.53%
	Average Precision	0.5840	0.5509	0.5799	-5.67%	-0.70%
name_calling_dumb_lazy	Balanced Accuracy	0.6369	0.6249	0.6387	-1.88%	+0.28%
	ROC-AUC	0.6938	0.6686	0.6894	-3.63%	-0.63%
	Average Precision	0.5177	0.4894	0.5085	-5.47%	-1.78%
hit_face_head_ears	Balanced	0.6399	0.6139	0.6284	-4.06%	-1.80%

	Accuracy					
	ROC-AUC	0.6985	0.6534	0.6718	-6.46%	-3.82%
	Average Precision	0.6907	0.6428	0.6659	-6.93%	-3.59%
hand_arm_leg	Balanced Accuracy	0.6275	0.5950	0.6141	-5.18%	-2.14%
	ROC-AUC	0.6738	0.6331	0.6584	-6.04%	-2.29%
	Average Precision	0.5431	0.5111	0.5332	-5.89%	-1.82%

Table F1: LOCO Robustness Analysis: Feature Importance Validation. Performance Comparison Across All Disciplinary Outcomes

Metric	Avg. Drop Without "Belief Physical Discipline"	Avg. Drop Without "Aggressive Behavior"
Balanced Accuracy	-3.25%	-1.09%
ROC-AUC	-4.47%	-2.04%

Average Precision	-4.73%	-1.70%

Table F2: Average Performance Drops Across All Outcomes

Appendix G

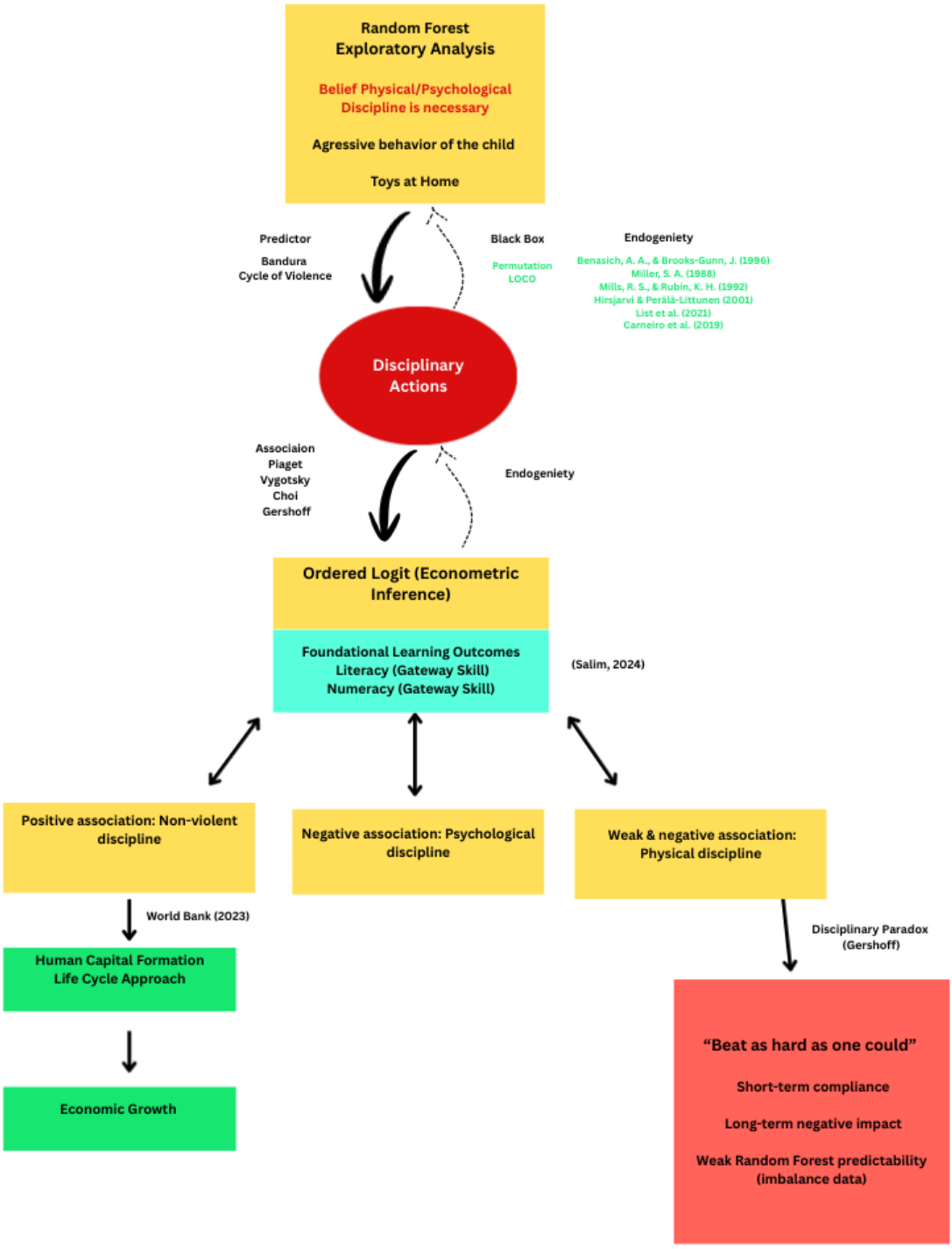

Random Forest
Exploratory Analysis
Belief Physical/Psychological Discipline is necessary
Agressive behavior of the child
Toys at Home
Predictor
Bandura
Cycle of Violence
Black Box
Permutation
LOCO
Endogeniety
Benasich, A. A., & Brooks-Gunn, J. (1996)
Miller, S. A. (1988)
Mills, R. S., & Rubin, K. H. (1992)
Hirsjarvi & Perälä-Littunen (2001)
List et al. (2021)
Carneiro et al. (2019)
Disciplinary Actions
Associaion
Piaget
Vygotsky
Choi
Gershoff
Endogeniety
Ordered Logit (Econometric Inference)
Foundational Learning Outcomes
Literacy (Gateway Skill)
Numeracy (Gateway Skill)
(Salim, 2024)
Positive association: Non-violent discipline
Negative association: Psychological discipline
Weak & negative association: Physical discipline
World Bank (2023)
Human Capital Formation
Life Cycle Approach
Economic Growth
Disciplinary Paradox
(Gershoff)
"Beat as hard as one could"
Short-term compliance
Long-term negative impact
Weak Random Forest predictability
(imbalance data)

Figure G: Hypothesized pathways linking caregiver beliefs, disciplinary practices, and child outcomes. The figure reflects evidence-based assumptions and serves as a reference for future studies.